

Exact Structure and Resource Bounds in Deterministic Pushdown Automata for Binary Counting Languages

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Abstract

We study the exact shortest-completion structure exposed by a deterministic pushdown automaton for

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\}.$$

Its end-of-input completion operation, zero-cancellation transition, and bounded stack-visibility property are shown to be manifestations of a single minimum-completion demand. Reachable left configurations $(L, 0^k)$ form an explicit ladder whose shortest continuations are exactly $0^i 1 0^{k+2-i}$, so completion and maximal cancellation are opposite boundary paths and adjacent choices form commuting diamonds in the actual configuration graph. The complete set of shortest first actions is determined by the control state and two visible stack symbols, and this depth is tight.

We extend the analysis to

$$L_{p,q} = \{w : qN_1(w) = pN_0(w)\}, \quad \gcd(p, q) = 1.$$

Every residual has a unique minimum demand (x_r, y_r) , while all shortest completions form the rectangle $[0, x_r] \times [0, y_r]$ and have the same Parikh vector. This yields exact counts of shortest completions and commuting squares, an exact geodesic-word growth formula, and canonical visibility depth $\max(p, q)$. We give both a four-symbol normalized residual-configuration conjugacy and an ε -free three-symbol raw conjugacy with $\max(p, q)$ states. Within permanent-bottom raw-conjugacy models, one work symbol has exact state complexity $p + q$, while the homogeneous-tail regime with at least two work symbols has exact state complexity $\max(p, q)$. The same three-symbol lower bound is proved for arbitrary encodings at ratios $(2, 1)$ and $(3, 1)$, and by a finite computer-assisted certificate at $(3, 2)$. Finally, a scoped phase-local representation model satisfies $|Q|g^\lambda \geq BH(B) \geq 2\sqrt{pqB}$. No unrestricted DPDA optimality claim is made.

1 Introduction

The language

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\}$$

can be recognized deterministically with counter-like memory by tracking the signed difference $N_1(w) - 2N_0(w)$. Deterministic one-counter models and signed counting mechanisms are classical [1, 2, 3]. Recognition itself is therefore not the question of interest here.

The motivating object of this paper is instead a concrete deterministic pushdown automaton whose stack behavior is more structured than a simple recognition counter. After a prefix has been read, its control state and stack jointly describe what is still required to reach the counting equality. If the input ends, a special completion operation converts this distributed information into a canonical shortest continuation. During ordinary input processing, the same machine

contains a left-state transition $0, 0/\varepsilon$ that consumes an input 0 while removing a stored 0. A third feature of the design is that a qualitative query about its completion stack can be answered by inspecting only the first two stack symbols, although the stack may be arbitrarily long.

These three mechanisms motivate the paper:

completion	cancellation	bounded stack visibility.
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At first sight they address different concerns. Completion is post-input behavior, cancellation occurs online, and bounded stack inspection is an information-access property. The main observation is that all three are governed by one common object: the *minimum completion demand*.

For a processed prefix w , define $\rho(w) = N_1(w) - 2N_0(w)$. If a continuation contains x zeros and y ones, then it completes the prefix exactly when $2x - y = \rho(w)$. Among all nonnegative solutions there is a unique pair minimizing $x + y$; call it $\mu(\rho) = (x_\rho, y_\rho)$. This pair determines how many zeros and ones every shortest completion must contain, but not their order. The demand is unique; the ordering of a shortest completion need not be.

The paper deliberately proceeds from the machine to the general theory. We first place the two formal variants of the original 2:1 DPDA in view and derive the reachable-store and residual invariants directly from their displayed transitions. The same configurations then yield the shortest-completion ladder, completion–cancellation duality, and the exact two-symbol local decoder. Only after those machine-level facts are established do we pass to

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}, \quad \gcd(p,q) = 1,$$

where the ladder becomes a shortest-completion rectangle and the residual coordinate supports explicit compact DPDAs and scoped state–alphabet resource bounds.

Contributions. The paper develops five linked results. First, the motivating DPDA realizes an exact shortest-completion ladder in its ordinary configuration graph, with cancellation and end-of-input completion as extremal schedules. Second, the complete set of shortest first actions is determined exactly by the control state and two visible stack symbols, and this depth is tight. Third, for every coprime p, q , the machine-level picture extends to residual-indexed shortest-completion rectangles and to explicit four- and three-symbol deterministic realizations. Fourth, the permanent-bottom raw-conjugacy models admit exact state–alphabet bounds, including $p + q$ states with one work symbol, $\max(p, q)$ states in the homogeneous-tail two-work-symbol regime, and tight arbitrary-encoding three-symbol lower bounds for $(2, 1)$, $(3, 1)$, and $(3, 2)$. Fifth, the phase-local representation model satisfies the scoped capacity inequality $|Q|g^\lambda \geq BH(B)$. The novelty claim is correspondingly narrow: the contribution is this proved family-specific synthesis and its exact scoped bounds, not the individual classical ingredients; section 3 states the literature boundary explicitly.

Relation to the companion manuscript. A companion manuscript studies completion frontiers, quotient geometry, and a compact normalized construction for the same counting family [11]. Because both papers start from the author’s original 2023 2:1 design, the two formal DPDA diagrams in figures 1 and 2 and the basic residual viewpoint are intentionally shared. The present manuscript restates every definition and construction needed for its arguments and does not rely on an unproved result from the companion paper. Its distinct contribution is the configuration-level shortest-completion ladder and commuting diamonds, the exact local shortest-action decoder, the general shortest-completion rectangles and geodesic-word growth law, the raw-defect classification, the three-symbol raw conjugacy, the exact scoped state–alphabet bounds, the ratio-specific arbitrary-encoding lower bounds, and the phase-local capacity theorem.

2 The motivating completion-aware DPDA

The 2:1 machine was developed by the author in 2023, before the present generalization and proof analysis. We therefore refer to it as the *original 2023 design*. The purpose of this section is to retain that design as the starting object while separating ordinary recognition, acceptance of the empty word, and end-of-input completion.

2.1 Original and formal presentations

The two automata in figures 1 and 2 implement the same center/left/right stack logic. They differ only in whether the empty input is accepted. The notation

$$a, A/\gamma$$

means: read input symbol a , replace stack top A by γ , and move along the indicated arc. The symbol Z_0 is the bottom marker and ε denotes the empty word.

The completion rules are displayed in dashed callout boxes rather than as ordinary graph edges. The dashed shape, not color alone, distinguishes them in grayscale. They are enabled only when the right endmarker $\dashv$ is the next unread symbol. Formally, each such rule can be directed to a dedicated completion sink q_{comp} , omitted from the drawings for clarity. This point is essential: replacing the original completion rules by ordinary ε -moves would conflict with input-consuming transitions from the same state/top pair and would violate the standard DPDA determinism condition. With $\dashv$, no such conflict occurs.

How to read the diagrams. The center state C carries the ordinary cancellation logic. The side states L and R are transient mismatch modes: they do not represent a second independent counter, but a finite correction to the meaning of the stack. Solid arrows process 0 and 1; the dashed blue boxes list only the rules that become available after the binary input has ended. The two figures have identical center/left/right behavior. Their only intended difference is whether the empty input is accepted.

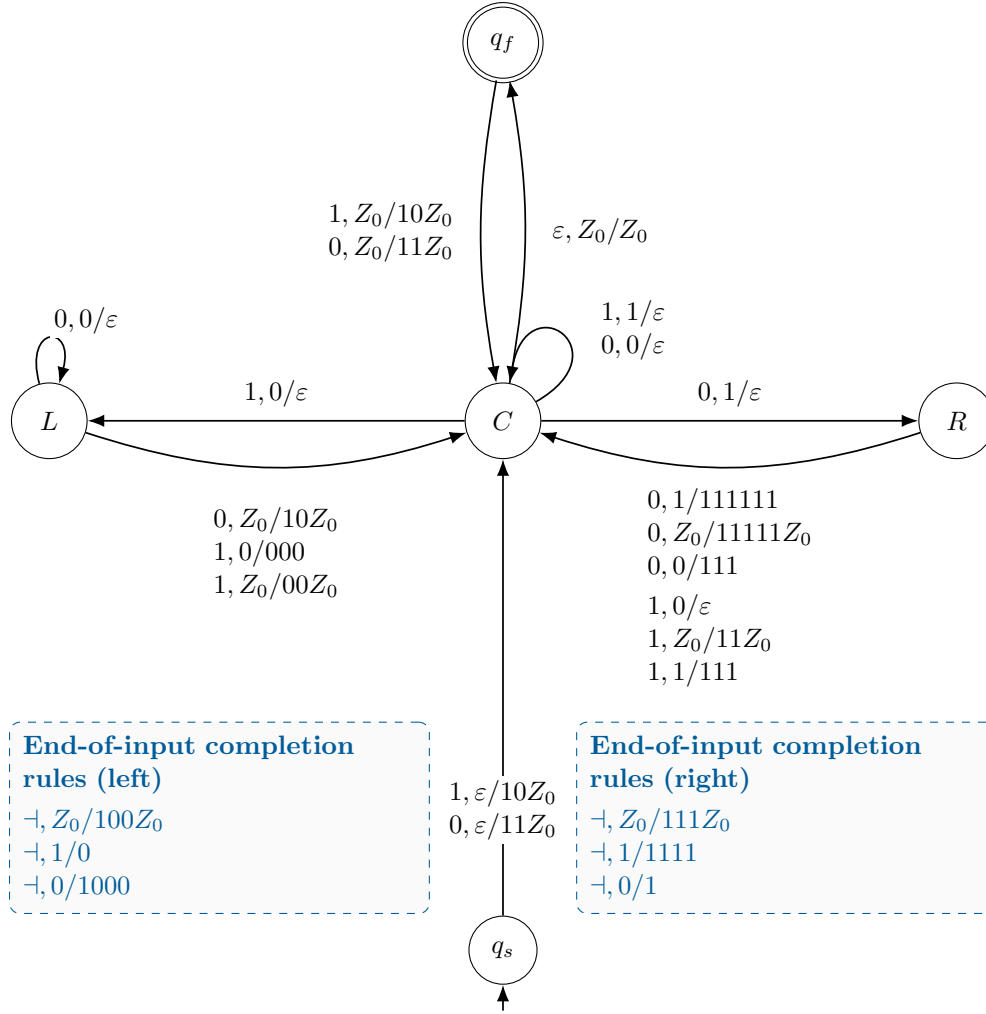


Figure 1: Formalized redrawing of the original 2:1 DPDA variant with the empty input excluded. Solid arrows are ordinary recognition transitions. Dashed callout boxes list the completion rules enabled only by the right endmarker $\neg$ after the binary input has been exhausted; the same distinction remains visible in grayscale. The left-state rule $\neg, 1/0$ is retained for fidelity to the original completion list, but its source condition is unreachable because every reachable left-state stack belongs to 0^* . The source design is archived at <https://github.com/alperenbutun/dpda-ones-double-zeros>.

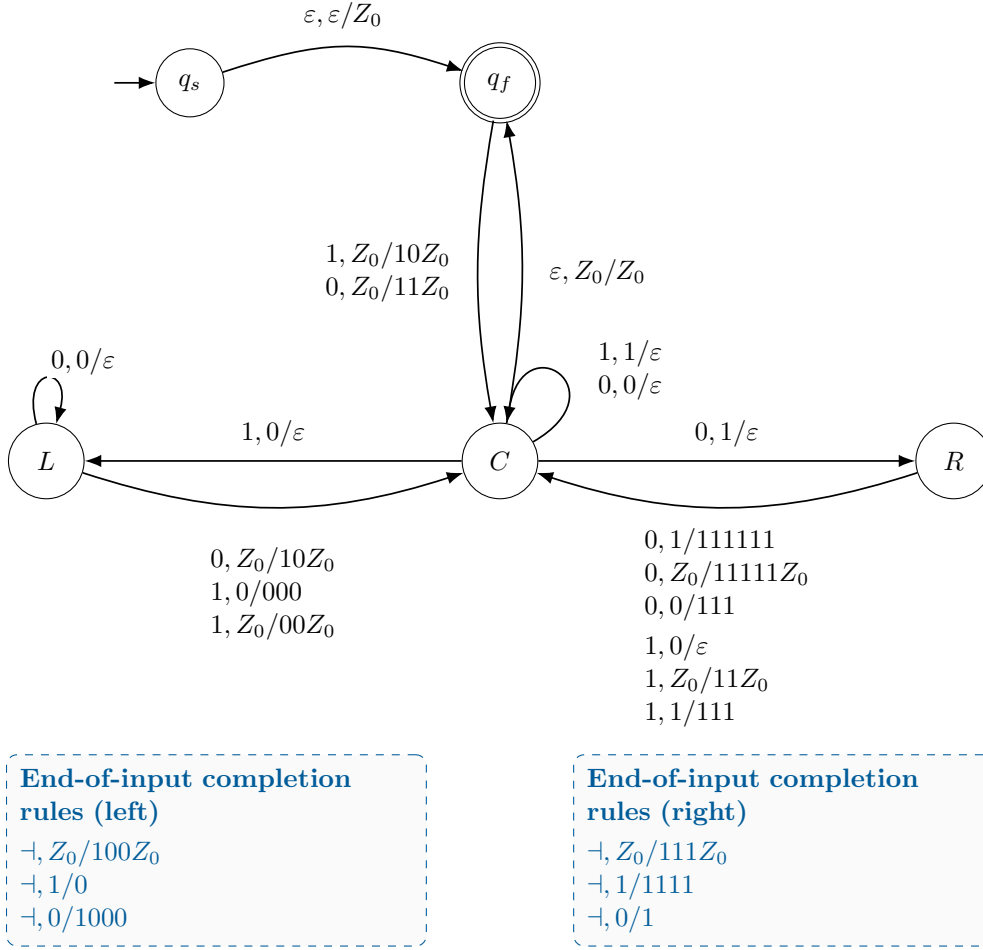


Figure 2: Formalized redrawing of the original 2:1 DPDA variant with the empty input included. The center/left/right recognition logic and the dashed end-of-input completion callouts are the same as in figure 1; the initialization/final-state structure additionally accepts ε . Hence this variant recognizes the standard language $L_{2,1}$, while the previous variant recognizes $L_{2,1} \setminus \{\varepsilon\}$. The source design is archived at <https://github.com/alperenbutun/dpda-ones-twice-zeros>.

The analysis below starts from these two formal variants. We first separate recognition from end-of-input completion, project away the initialization/acceptance wrapper, and derive the reachable-store and residual invariants directly from the displayed transitions. Only after that machine-level analysis do we pass to the general $L_{p,q}$ constructions and resource bounds.

2.2 Formal right-endmarker completion semantics

The archived drawings use dotted ε -labels for the completion rules. Those labels record the original design notation; they are *not* interpreted as ordinary ε -moves in the machine analyzed here. At binary input exhaustion we use the deterministic post-input operator shown in table 1. Equivalently, each row may be implemented by a transition labeled by the right endmarker $\neg$ into a terminal normalization mode. This keeps completion disjoint from ordinary 0/1 recognition moves.

Source	top	replacement	normalized view
L	Z_0	$100Z_0$	C
L	1	0	C (unreachable source)
L	0	1000	C
R	Z_0	$111Z_0$	C
R	1	1111	C
R	0	1	C

Table 1: Formal post-input completion semantics. Every row is enabled only when $\neg$ is the next unread symbol. The operation exposes a canonical completion witness; it does not accept a nonmember input. Membership of the already-read word is still determined by zero residual.

This distinction also separates two roles that are easy to conflate: *recognition* asks whether the residual of the completed input is zero, whereas *completion* asks for a shortest suffix that would make it zero. The latter is semantic postprocessing and is used below to expose what the stack stores.

2.3 Core projection and solid transition semantics

Write a physical core stack as SZ_0 , where $S \in \{0, 1\}^*$ is the logical stack. The wrapper states are projected to the zero-residual center view by

$$\pi(q_s, \varepsilon) = (C, \varepsilon), \quad \pi(q_f, Z_0) = (C, \varepsilon),$$

and for $q \in \{C, L, R\}$,

$$\pi(q, SZ_0) = (q, S).$$

Whenever we speak of “the core configuration after a processed prefix”, this projection is understood. In particular, the two wrapper moves that resume reading from the balanced configuration induce the projected rules on top Z_0 shown in the first two rows of table 2.

source	input	top	replacement	target
C	1	Z_0	$10Z_0$	C
C	0	Z_0	$11Z_0$	C
C	1	0	ε	L
C	0	1	ε	R
C	1	1	ε	C
C	0	0	ε	C
L	0	0	ε	L
L	0	Z_0	$10Z_0$	C
L	1	0	000	C
L	1	Z_0	$00Z_0$	C
R	0	1	111111	C
R	0	Z_0	11111 Z_0	C
R	0	0	111	C
R	1	0	ε	C
R	1	Z_0	11 Z_0	C
R	1	1	111	C

Table 2: Solid recognition semantics after core projection. The first two rows are wrapper-induced projected moves from the balanced configuration; all remaining rows are the solid center/left/right rules visible in the diagrams. The table is the transition relation used in the machine-level proofs below.

2.4 Reachable stores and state potential

With the projection above, the semantic core consists only of C, L, R and logical stacks in $\{0, 1\}^*$.

Proposition 2.1 (Reachable-store normal form). *Every reachable core configuration (q, S) satisfies*

$$S \in \mathcal{S}_q, \quad \mathcal{S}_C = 0^* \cup 1^* \cup \{10\}, \quad \mathcal{S}_L = 0^*, \quad \mathcal{S}_R = 1^* \cup \{0\}.$$

In particular, no reachable left-state configuration has top symbol 1.

Proof. Induct on consumed input length. Center loops cancel matching symbols and preserve the unary families; center-to-side mismatch transitions enter only the displayed side families; the left loop $0, 0/\varepsilon$ preserves 0^* ; and every side-to-center transition returns to $0^*, 1^*$, or the exceptional center word 10. No transition enters L with top 1. $\square$

Define

$$D(S) = 2N_0(S) - N_1(S), \quad \Phi(C) = 0, \quad \Phi(L) = 3, \quad \Phi(R) = -3.$$

Proposition 2.2 (State-potential invariant). *If prefix w leaves the core configuration (q, S) , then*

$$\boxed{\rho(w) = D(S) + \Phi(q)}.$$

Proof. The claim holds at the projected empty configuration (C, ε) . A direct check of every row of table 2 shows that the change in $D(S) + \Phi(q)$ equals $+1$ on input 1 and -2 on input 0, exactly matching the change in ρ . This includes the wrapper-induced projected moves from top Z_0 and the left cancellation $0, 0/\varepsilon$, for which both quantities decrease by 2. $\square$

For a reachable core configuration $c = (q, S)$, we henceforth write

$$\rho(c) := D(S) + \Phi(q), \quad \mu(c) := \mu(\rho(c)), \quad \Delta(c) := \Delta(\rho(c))$$

whenever the completion-demand notation has been introduced. This is only shorthand: the residual remains the semantic coordinate.

2.5 Worked completion example: the prefix 111

Using the empty-input-excluded presentation and suppressing Z_0 from the logical stack notation,

$$(C, 111 \dashv, \varepsilon) \vdash (C, 11 \dashv, 10) \vdash (C, 1 \dashv, 0) \vdash (L, \dashv, \varepsilon).$$

Here $\rho(111) = 3$. The empty logical stack does not mean zero residual, because $\Phi(L) = 3$. The endmarker rule $\dashv, Z_0/100Z_0$ materializes the completion word 100. Indeed, every completion with x zeros and y ones must satisfy $2x - y = 3$, whose unique minimum nonnegative solution is $(x, y) = (2, 1)$. Thus the shortest completions are 100, 010, 001, and the machine selects 100 as the canonical ordering.

Lemma 2.3 (Canonical end-of-input materialization). *Let $c = (q, S)$ be any reachable core configuration, and let $\widehat{S}(c)$ be the logical center stack obtained by applying the unique end-of-input completion rule when $q \in \{L, R\}$ and doing nothing when $q = C$. Then*

$$\boxed{\widehat{S}(c) = \begin{cases} 0^m, & \rho(c) = 2m, \quad m \in \mathbb{N}_0, \\ 10^{m+1}, & \rho(c) = 2m + 1, \quad m \in \mathbb{N}_0, \\ 1^m, & \rho(c) = -m, \quad m \geq 1. \end{cases}}$$

In particular,

$$D(\widehat{S}(c)) = \rho(c),$$

and, interpreted as a continuation word, $\widehat{S}(c)$ is the canonical shortest completion for the represented residual.

Proof. Use the reachable-store families of theorem 2.1. If $q = C$, then S is already one of 0^m , 1^m , or 10 , and the displayed formula follows directly from $\rho(c) = D(S)$. If $c = (L, 0^k)$, then $\rho(c) = 2k + 3$; the Z_0 completion rule for $k = 0$ writes 100 , while for $k \geq 1$ the rule on top 0 replaces the first 0 by 1000 , yielding 10^{k+2} in both cases. If $c = (R, 1^k)$, including $k = 0$, then $\rho(c) = -(k + 3)$ and the right completion rule yields 1^{k+3} ; the exceptional configuration $(R, 0)$ has residual -1 and normalizes to 1 . The left rule requiring top 1 is unreachable by theorem 2.1. Each displayed word has the unique minimum nonnegative Parikh pair solving $2x - y = \rho(c)$, so it is shortest. $\square$

3 Related work and scope of contribution

Counters and shortest paths. Deterministic one-counter automata are classical [1], and modern work gives tight bounds for shortest paths between arbitrary configurations in one-counter transition systems [4]. Our residual graph is deliberately elementary relative to that literature: the point is not a new shortest-path formalism, but that the shortest paths can be read back into the stack operations of the motivating DPDA.

Shortest-edge marking on pushdown graphs. Carayol and Serre consider deterministic real-time pushdown graphs with designated final vertices and mark every outgoing edge lying on some shortest path to a final vertex [8]. This is the closest prior concept to our GeoFirst sets. The distinction in the present paper is exact specialization rather than generalization: on the reachable configurations of one concrete completion-aware DPDA, the marked-edge role has an explicit arithmetic decoder, is exactly two-local, and forms a complete ladder of configuration-level commuting diamonds on the left branch.

Commutation and normal forms. Trace monoids identify words modulo repeated commutation of independent adjacent letters [9]; geodesic normal forms are also well established in group-theoretic language theory [6]. Consequently, the rewrite $01 \rightarrow 10$ and its inversion-count termination argument are not presented as generic rewriting novelty. Their role here is to identify the machine-selected canonical completion as one boundary representative of an explicitly realized shortest-computation interval.

Stores and descriptive resources. Store languages formalize the state-plus-store configurations that arise during accepting computations and are classical for pushdown automata [5]. Descriptive complexity also depends strongly on the number of stack symbols: Masopust establishes stack-alphabet hierarchies for restricted deterministic pushdown models [7], while Martynova gives exact simultaneous state and stack-symbol determinization bounds for input-driven pushdown automata [10]. These results rule out any broad claim that state/stack tradeoffs are new. Our resource theorem asks a different, representation-scoped question by adding an explicit semantic visibility depth λ and counting modular completion phases that must remain locally distinguishable.

Position of the present results. Accordingly, our contribution is not that geodesics, commuting actions, stack visibility, real-time pushdown computation, or descriptonal tradeoffs are individually new. It is the explicit synthesis obtained for this completion-aware family: exact minimum-demand descent, a raw configuration-level ladder in the motivating machine, a tight local decoder for the shortest-edge role, a canonical materialization map, an exact normalized configuration conjugacy for the four-symbol compact DPDA, a finite and tight boundary-defect classification for that fixed encoding, an exact $p + q$ state theorem for one-work-symbol raw conjugacy, a three-symbol raw-conjugate realization with the $\max(p, q)$ control-state budget and a corresponding fixed-budget alphabet lower bound, exact arbitrary-encoding three-symbol state bounds for the ratios $(2, 1)$, $(3, 1)$, and $(3, 2)$ within the permanent-bottom raw-conjugacy model, and a modular phase-capacity bound. The general $L_{p,q}$ rectangle is the arithmetic object connecting these machine-specific facts; its lattice-path content is intentionally not advertised as an isolated invention.

4 Minimum completion demand and geodesic descent

Definition 4.1 (Minimum completion demand). For residual $r \in \mathbb{Z}$, let

$$\mu(r) = (x_r, y_r)$$

be the unique nonnegative solution minimizing $x + y$ subject to $2x - y = r$, and let

$$\Delta(r) = x_r + y_r.$$

Define $T_0(r) = r - 2$ and $T_1(r) = r + 1$. A symbol $a \in \{0, 1\}$ is *geodesic at r* if $\Delta(T_a(r)) = \Delta(r) - 1$.

For $L_{2,1}$,

$$\mu(2m) = (m, 0), \quad \mu(2m + 1) = (m + 1, 1), \quad \mu(-m) = (0, m) \quad (m \geq 1).$$

Lemma 4.2 (Minimum-demand geodesic descent). *Let $\mu(r) = (x_r, y_r)$. If $x_r > 0$, then*

$$\mu(r - 2) = (x_r - 1, y_r), \quad \Delta(r - 2) = \Delta(r) - 1.$$

If $y_r > 0$, then

$$\mu(r + 1) = (x_r, y_r - 1), \quad \Delta(r + 1) = \Delta(r) - 1.$$

Consequently

$$\text{GeoFirst}(r) = \{0 : x_r > 0\} \cup \{1 : y_r > 0\}.$$

Proof. If $x_r > 0$, then $(x_r - 1, y_r)$ satisfies $2x - y = r - 2$. If a shorter solution existed for $r - 2$, adding one zero would give a completion of r shorter than $\mu(r)$, a contradiction. The argument for 1 is symmetric. $\square$

For $c_k = (L, 0^k)$, theorem 2.2 gives

$$\rho(c_k) = 2k + 3, \quad \mu(c_k) = (k + 2, 1), \quad \Delta(c_k) = k + 3.$$

The transition $(L, 0^k) \xrightarrow{0} (L, 0^{k-1})$ therefore maps $(k + 2, 1)$ to $(k + 1, 1)$ and decreases Δ by one. The boundary transitions

$$(L, \varepsilon) \xrightarrow{0} (C, 10) \xrightarrow{0} (R, 0) \xrightarrow{1} (C, \varepsilon)$$

continue the same exact descent.

5 Operational geodesic structure of the motivating DPDA

The ladder sources used below are genuine reachable configurations, not merely syntactically legal stack patterns. From the projected empty configuration, table 2 gives $111 \mapsto (L, \varepsilon)$, and two further 1's map $(L, 0^k)$ to $(L, 0^{k+1})$ through the center. Hence, by induction,

$$\boxed{1^{2k+3} \text{ reaches } (L, 0^k) \quad (k \geq 0).}$$

Theorem 5.1 (Exact geodesic ladder). *For every $k \geq 0$, define*

$$\Psi_k(i, 0) = \begin{cases} (L, 0^{k-i}), & 0 \leq i \leq k, \\ (C, 10), & i = k+1, \\ (R, 0), & i = k+2, \end{cases}$$

and

$$\Psi_k(i, 1) = (C, 0^{k+2-i}), \quad 0 \leq i \leq k+2.$$

Then every horizontal edge of $[0, k+2] \times [0, 1]$ is realized by a solid 0-transition, every vertical edge by a solid 1-transition, and every monotone source-to-target path is a shortest zero-residual continuation in the semantic core from $(L, 0^k)$ to (C, ε) . Under the ordinary final-state wrapper, ending the input there yields acceptance. Hence the shortest continuations are exactly

$$\boxed{0^i 1 0^{k+2-i}, \quad 0 \leq i \leq k+2.}$$

There are exactly $k+3$ such words.

Proof. For lower horizontal edges, use the left cancellation loop while $i < k$, then $(L, \varepsilon) \xrightarrow{0} (C, 10)$, then $(C, 10) \xrightarrow{0} (R, 0)$. Upper horizontal edges are center cancellations $(C, 0^m) \xrightarrow{0} (C, 0^{m-1})$. For vertical rungs: when $0 \leq i < k$, the rule $1, 0/000$ gives $(L, 0^{k-i}) \xrightarrow{1} (C, 0^{k+2-i})$; at $i = k$, the rule $1, Z_0/00Z_0$ gives $(L, \varepsilon) \xrightarrow{1} (C, 00)$; at $i = k+1$, the center rule $1, 1/\varepsilon$ gives $(C, 10) \xrightarrow{1} (C, 0)$; and at $i = k+2$, the right rule $1, 0/\varepsilon$ gives $(R, 0) \xrightarrow{1} (C, \varepsilon)$. Every path consumes the unique minimum demand $(k+2, 1)$, hence is shortest by theorem 4.2. $\square$

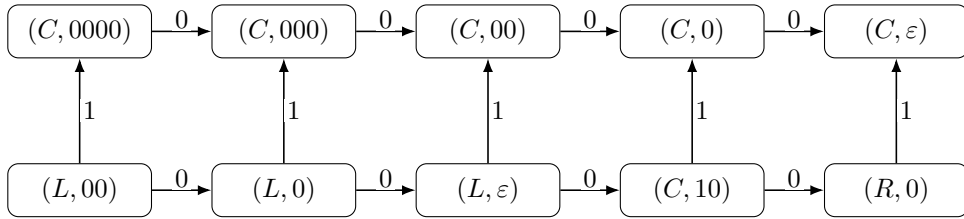


Figure 3: The exact ladder for $k = 2$. The source $(L, 00)$ has residual 7 and minimum demand $(4, 1)$. Its five shortest continuations are 10000, 01000, 00100, 00010, 00001. The upper boundary is canonical completion; the lower boundary postpones the unique required 1 maximally.

Lemma 5.2 (Local commuting diamonds). *At every square of the ladder,*

$$\boxed{\delta^*(c, 01) = \delta^*(c, 10).}$$

Proof. For interior left configurations $(L, 0^m)$ with $m \geq 1$,

$$(L, 0^m) \xrightarrow{0} (L, 0^{m-1}) \xrightarrow{1} (C, 0^{m+1})$$

and

$$(L, 0^m) \xrightarrow{1} (C, 0^{m+2}) \xrightarrow{0} (C, 0^{m+1}).$$

At (L, ε) both 01 and 10 reach $(C, 0)$; at $(C, 10)$ both reach (C, ε) . These are exactly the boundary squares in theorem 5.1. $\square$

Corollary 5.3 (Geodesic Parikh synchronization). *Let u and v be prefixes of shortest continuations from $(L, 0^k)$. If they have the same consumed Parikh pair, then*

$$\delta^*((L, 0^k), u) = \delta^*((L, 0^k), v).$$

In particular every shortest continuation ends in (C, ε) .

Proof. Repeatedly exchange adjacent 01 and 10 factors using theorem 5.2. Within the ladder, the consumed Parikh pair uniquely determines the grid coordinate $\Psi_k(i, j)$. $\square$

Corollary 5.4 (Completion–cancellation boundary duality). *For $(L, 0^k)$, the canonical completion 10^{k+2} and maximal-cancellation continuation $0^{k+2}1$ are the two extremal boundary paths of the same geodesic ladder.*

Define the *commutation potential*

$$\Xi(r) = x_r y_r.$$

For $(L, 0^k)$, $\Xi = k + 2$. This equals the ladder’s number of elementary diamonds, the maximum number of consecutive optimal zeros before the final 1 is geodesically forced, and the adjacent-swap distance between 10^{k+2} and $0^{k+2}1$.

Corollary 5.5 (Cancellation–diamond identity). *For every reachable left configuration,*

$$\boxed{\Xi = \text{diamond count} = \text{cancellation horizon} = \text{extremal swap distance}.}$$

6 Exact local geodesic visibility

For a reachable core configuration c , let

$$\mathcal{G}(c) = \text{GeoFirst}(\rho(c)).$$

The reachable roles are

$$(C, \varepsilon) \mapsto \emptyset, \quad (C, 0^m) \mapsto \{0\}, \quad (C, 1^m) \mapsto \{1\}, \quad (C, 10) \mapsto \{0, 1\},$$

with every reachable $(L, 0^m)$ mapping to $\{0, 1\}$ and every reachable right-state configuration mapping to $\{1\}$.

For $d \geq 0$, let $\text{pref}_d^\perp(S)$ be the first d stack symbols padded by $\perp$, and define

$$\text{Obs}_d(q, S) = (q, \text{pref}_d^\perp(S)).$$

Theorem 6.1 (Exact depth-two geodesic locality). *For the motivating DPDA:*

(i) *there is a finite decoder D_2 such that*

$$\mathcal{G}(c) = D_2(\text{Obs}_2(c))$$

for every reachable core configuration;

(ii) depth one is insufficient, witnessed by $(C, 1)$ and $(C, 10)$;

(iii) after end-of-input normalization to a center stack $\widehat{S}(c)$,

$$\boxed{\text{alph}(\widehat{S}(c)) = \mathcal{G}(c)};$$

(iv) no fixed finite depth d determines the full minimum-demand vector $\mu(c)$ on all reachable configurations.

Thus the complete qualitative shortest-action geometry is exactly two-local, whereas the quantitative demand magnitude is not fixed-depth local.

Proof. By theorem 2.1, the reachable center stacks are ε , 0^m , 1^m , and 10 . At C , the observations $\perp\perp$, $0\perp/00$, $1\perp/11$, and 10 decode respectively to $\emptyset$, $\{0\}$, $\{1\}$, and $\{0, 1\}$. The state alone fixes the reachable left role $\{0, 1\}$ and right role $\{1\}$. Depth one fails because $(C, 1)$ and $(C, 10)$ share state and top symbol but have roles $\{1\}$ and $\{0, 1\}$. By theorem 2.3, completion normalization produces the canonical shortest completion for the represented residual; its symbol support is $\{0 : x_r > 0\} \cup \{1 : y_r > 0\} = \mathcal{G}(c)$ by theorem 4.2. Finally, the center configurations $(C, 0^m)$ are reachable for arbitrarily large m : $(C, 00)$ is reached after the prefix 1111 , and for every $m \geq 2$ the input block 11 maps $(C, 0^m)$ to $(C, 0^{m+1})$. Hence, for any fixed d , choose distinct reachable $m, n \geq \max\{d, 2\}$. The two configurations have the same state and top- d observation but minimum demands $(m, 0)$ and $(n, 0)$. $\square$

The theorem gives a precise interpretation of the original bounded-search intuition: two stack symbols do not reconstruct the stack; they recover whether the current shortest-completion geometry is balanced, a one-way corridor, or a commuting fork.

7 General shortest-completion geometry for $L_{p,q}$

Fix coprime positive integers p, q and define

$$\rho_{p,q}(w) = qN_1(w) - pN_0(w).$$

A continuation with x zeros and y ones completes residual r exactly when

$$px - qy = r.$$

Proposition 7.1 (Unique minimum Parikh demand). *For every $r \in \mathbb{Z}$, there is a unique nonnegative pair $\mu(r) = (x_r, y_r)$ minimizing $x + y$ subject to $px - qy = r$. All nonnegative solutions are*

$$(x_r + qk, y_r + pk), \quad k \geq 0.$$

Proof. Bézout's identity gives at least one integer solution (x_0, y_0) of $px - qy = r$. Every integer solution is

$$(x_0 + qt, y_0 + pt), \quad t \in \mathbb{Z},$$

because $\gcd(p, q) = 1$. For sufficiently large t both coordinates are nonnegative, so feasible nonnegative solutions exist. The feasible values of t form a ray $t \geq t_{\min}$, and increasing t by one increases the total length by $p + q$. Hence the first feasible pair is uniquely minimal, and every other nonnegative solution is obtained from it by adding (q, p) a nonnegative number of times. $\square$

Set $\Delta(r) = x_r + y_r$. Let $\mathcal{R}_{p,q}$ be the directed residual graph with vertex set $\mathbb{Z}$ and edges

$$r \xrightarrow{0} r - p, \quad r \xrightarrow{1} r + q.$$

Lemma 7.2 (General geodesic descent and bounded coordinate). *For $\mu(r) = (x_r, y_r)$,*

$$0 \in \text{GeoFirst}_{p,q}(r) \iff x_r > 0, \quad 1 \in \text{GeoFirst}_{p,q}(r) \iff y_r > 0.$$

More precisely, if $x_r > 0$ then $\mu(r - p) = (x_r - 1, y_r)$, and if $y_r > 0$ then $\mu(r + q) = (x_r, y_r - 1)$. In addition,

$$r > 0 \implies x_r > 0 \text{ and } 0 \leq y_r < p, \quad r < 0 \implies y_r > 0 \text{ and } 0 \leq x_r < q.$$

Proof. The descent argument is the same minimality argument as in theorem 4.2. For the bounded coordinate, assume $r > 0$. Then $x_r > 0$. If $y_r \geq p$, the equation $px_r - qy_r = r > 0$ forces $x_r > q$, so $(x_r - q, y_r - p)$ is a shorter nonnegative solution, contradicting minimality. Hence $y_r < p$. The case $r < 0$ is symmetric. $\square$

Theorem 7.3 (Rectangular shortest-completion geometry). *Let $\mu(r) = (x_r, y_r)$. The map*

$$(i, j) \mapsto r - pi + qj$$

is injective on $[0, x_r] \times [0, y_r]$. Hence the directed shortest-path interval from r to 0 in $\mathcal{R}_{p,q}$ is exactly the rectangular grid

$$\boxed{[0, x_r] \times [0, y_r]}.$$

Consequently

$$\boxed{\text{GeoComp}(r) = \{z : N_0(z) = x_r, N_1(z) = y_r\}},$$

$$\boxed{|\text{GeoComp}(r)| = \binom{x_r + y_r}{x_r}}, \quad \boxed{\Xi(r) = x_r y_r},$$

where $\Xi(r)$ is the number of elementary commuting squares.

Proof. If $r - pi + qj = r - pi' + qj'$, then $p(i - i') = q(j - j')$, so $i - i' = qt$ and $j - j' = pt$. A nonzero t would force $x_r \geq q$ and $y_r \geq p$, making $(x_r - q, y_r - p)$ a shorter nonnegative solution of the same completion equation, contradicting theorem 7.1. Thus the map is injective. Every monotone path consumes exactly the minimum demand, and every shortest completion must have that Parikh vector. $\square$

Theorem 7.4 (Exact geodesic-word growth). *Let $G_{p,q}(n)$ be the number of binary words of length n that label a directed geodesic between their endpoints in $\mathcal{R}_{p,q}$. Then a word z is geodesic exactly when*

$$\boxed{N_0(z) < q \quad \text{or} \quad N_1(z) < p.}$$

Consequently

$$\boxed{G_{p,q}(n) = 2^n \quad (0 \leq n < p + q),}$$

while for $n \geq p + q$,

$$\boxed{G_{p,q}(n) = \sum_{i=0}^{q-1} \binom{n}{i} + \sum_{j=0}^{p-1} \binom{n}{j}}.$$

The first non-geodesic words occur at length $p + q$ and are exactly the $\binom{p+q}{q}$ orderings of the fundamental balanced Parikh vector (q, p) . If $(p, q) \neq (1, 1)$ and $M = \max(p, q)$, then

$$\boxed{G_{p,q}(n) \sim \frac{n^{M-1}}{(M-1)!}}.$$

For $(p, q) = (1, 1)$, one has $G_{1,1}(0) = 1$ and $G_{1,1}(n) = 2$ for every $n \geq 1$.

Proof. Let $x = N_0(z)$ and $y = N_1(z)$. The word z is shortest between its endpoints exactly when (x, y) is the minimum nonnegative solution for its displacement. By theorem 7.1, every other nonnegative solution differs by a nonnegative multiple of (q, p) . Hence (x, y) fails to be minimum exactly when $x \geq q$ and $y \geq p$, because then $(x - q, y - p)$ is a shorter nonnegative solution with the same displacement. This proves the displayed geodesic criterion.

If $n = x + y < p + q$, the two inequalities $x \geq q$ and $y \geq p$ cannot hold simultaneously, so all 2^n words are geodesic. For $n \geq p + q$, the geodesic regions $x < q$ and $y < p$ are disjoint. There are $\binom{n}{i}$ words with exactly i zeros and $\binom{n}{j}$ words with exactly j ones, giving the two binomial tails above. At the first excluded length $p + q$, the only nonminimum Parikh pair is (q, p) , so precisely its $\binom{p+q}{q}$ orderings are non-geodesic.

For a nontrivial coprime pair one has $p \neq q$. The larger of the two fixed binomial tails has largest index $M - 1$, and $\binom{n}{M-1} \sim n^{M-1}/(M-1)!$, while every other term has lower polynomial degree. The case $(1, 1)$ follows directly from the criterion: a nonempty geodesic word must be unary. $\square$

For $r > 0$, the canonical boundary geodesic is $C(r) = 1^{y_r} 0^{x_r}$; for $r < 0$, it is $C(r) = 0^{x_r} 1^{y_r}$. On the fixed shortest-completion class, the adjacent rewrite $01 \rightarrow 10$ for $r > 0$ (respectively $10 \rightarrow 01$ for $r < 0$) strictly decreases the appropriate inversion count and has $C(r)$ as unique normal form. The maximum normalization distance is $x_r y_r = \Xi(r)$.

Corollary 7.5 (Exact geodesic-fork classification). *For $r > 0$,*

$$\text{GeoFirst}_{p,q}(r) = \begin{cases} \{0\}, & p \mid r, \\ \{0, 1\}, & p \nmid r, \end{cases}$$

and for $r < 0$,

$$\text{GeoFirst}_{p,q}(r) = \begin{cases} \{1\}, & q \mid r, \\ \{0, 1\}, & q \nmid r. \end{cases}$$

At $r = 0$ the action set is empty.

Proof. For $r > 0$, theorem 7.2 gives $x_r > 0$ and $y_r \in \{0, \dots, p-1\}$, while the completion equation gives $q y_r \equiv -r \pmod{p}$. Since q is invertible modulo p , $y_r = 0$ iff $p \mid r$. The negative case is symmetric using $p x_r \equiv r \pmod{q}$ with $x_r \in \{0, \dots, q-1\}$. $\square$

Thus $(1, 1)$ has no forks; $p > 1, q = 1$ has positive-side forks only; $p = 1, q > 1$ has negative-side forks only; and $p, q > 1$ has forks on both sides. The motivating $(2, 1)$ machine is therefore one-sided for an arithmetic reason, not an accidental graphical one.

Because $C(r)$ contains exactly the symbol types whose demand coordinates are positive,

$$\boxed{\text{alph}(C(r)) = \text{GeoFirst}_{p,q}(r).}$$

Theorem 7.6 (Exact canonical geodesic-action visibility). *Let d be the least integer such that, for every residual r ,*

$$\text{alph}(C(r)) = \text{alph}(\text{pref}_d(C(r))).$$

Then

$$\boxed{d = \lambda_{\text{geo}}^{\text{canonical}}(p, q) = \max(p, q).}$$

Thus the complete geodesic first-action set is recoverable from exactly the same worst-case prefix depth.

Proof. For $r > 0$, write $C(r) = 1^{y_r}0^{x_r}$. By theorem 7.2, $y_r < p$. If both symbols occur, the first 0 appears by position $y_r + 1 \leq p$; if only one symbol occurs, its support is visible immediately. Hence depth p suffices on the positive side. Symmetrically, depth q suffices on the negative side, so $d \leq \max(p, q)$.

For the lower bound, suppose $p > 1$ and take $r = q$. By theorem 7.2, the minimum solution has $0 \leq y_r < p$, while $qy_r \equiv -q \pmod{p}$. Since q is invertible modulo p , this forces $y_r = p - 1$, and then the completion equation forces $x_r = q$. Thus

$$C(q) = 1^{p-1}0^q,$$

so depth $p - 1$ misses a symbol that occurs later and $d \geq p$. If $q > 1$, the symmetric congruence for residual $r = -p$ forces $(x_r, y_r) = (q - 1, p)$ and hence gives

$$C(-p) = 0^{q-1}1^p,$$

forcing $d \geq q$. The only case not covered by one of these witnesses is $p = q = 1$, where any nonzero one-letter canonical completion shows $d \geq 1$. Therefore $d \geq \max(p, q)$, completing the proof. $\square$

The companion completion-frontier analysis derives the same numerical depth from a frontier factorization viewpoint [11]; no result from that manuscript is needed for the proof above.

7.1 Lifting residual rectangles to normalized DPDA configurations

The rectangle theorem has so far been stated in the residual graph. We now show that it is realized exactly by the normalized configurations of an explicit compact DPDA. We use the right-endmarker paired-sign construction, reproducing the part of its encoding needed for the argument; the full transition schema is given in the companion construction paper [11].

Let

$$m = \max(p, q), \quad \Gamma = \{0, 1, X, Z_0\},$$

and use stable control states $R_0, \dots, R_{m-1}$. For $0 \leq s < p$ and $0 \leq t < q$, put

$$a_s = \begin{cases} 0, & s = 0, \\ s - p, & 1 \leq s < p, \end{cases} \quad b_t = \begin{cases} 0, & t = 0, \\ q - t, & 1 \leq t < q. \end{cases}$$

Define the stable encoding $\eta : \mathbb{Z} \rightarrow Q \times \Gamma^*$ by

$$\eta(0) = (R_0, Z_0).$$

If $r > 0$, let $s = r \bmod p$ and write uniquely

$$r = pk + a_s, \quad k \geq 1;$$

then

$$\eta(r) = (R_s, 0^k Z_0).$$

If $r < 0$, let $t = (-r) \bmod q$ and write uniquely

$$r = -qk + b_t, \quad k \geq 1;$$

then

$$\eta(r) = (R_t, 1^k X).$$

Although positive and negative roles may share a control state, their stack symbols and sentinels distinguish the signs, so η is injective.

For completeness, the semantic rules needed below are recorded here. For $0 \leq s < p$, set

$$s' = (s + q) \bmod p, \quad c_s = \frac{a_s + q - a_{s'}}{p},$$

and use

$$\delta(R_s, 0, 0) = (R_s, \varepsilon), \quad \delta(R_s, 1, 0) = (R_{s'}, 0^{c_s+1}).$$

When the last positive 0 is popped, the boundary rule is

$$\delta(R_s, \varepsilon, Z_0) = \eta(a_s) \quad (1 \leq s < p),$$

while $s = 0$ already leaves the zero configuration (R_0, Z_0) . For $0 \leq t < q$, set

$$t' = (t + p) \bmod q, \quad d_t = \frac{p - b_t + b_{t'}}{q},$$

and use

$$\delta(R_t, 1, 1) = (R_t, \varepsilon), \quad \delta(R_t, 0, 1) = (R_{t'}, 1^{d_t+1}),$$

with boundary normalization

$$\delta(R_t, \varepsilon, X) = \eta(b_t).$$

Finally, at zero,

$$\delta(R_0, 1, Z_0) = \eta(q), \quad \delta(R_0, 0, Z_0) = \eta(-p).$$

Here an expression $\delta(\cdot) = \eta(r')$ at a sentinel boundary means that the exposed sentinel is replaced by the stack word of $\eta(r')$ while control enters its state. These semantic rules are deterministic because binary input is disabled on the transient boundary pairs carrying the forced ε -move.

A *normalized input macro-step*

$$c \xRightarrow{a} c'$$

means: consume the binary symbol a , then execute the unique forced finite ε -chain, if any, until the next stable configuration in $\eta(\mathbb{Z})$ is reached. In the unbounded-write semantic schema this chain is needed only at sign/zero boundaries; in the push-at-most-two compilation it also completes a forced homogeneous suffix write. In both cases it is deterministic and consumes no additional input.

Theorem 7.7 (Normalized residual–configuration conjugacy). *For every $r \in \mathbb{Z}$,*

$$\boxed{\eta(r) \xRightarrow{0} \eta(r - p), \quad \eta(r) \xRightarrow{1} \eta(r + q).}$$

Consequently η is a label-preserving graph isomorphism from the residual graph $\mathcal{R}_{p,q}$ onto the reachable normalized stable configuration graph of the compact paired-sign DPDA. For every $u \in \{0, 1\}^$,*

$$\boxed{\widehat{\delta}^*(\eta(r), u) = \eta(r - pN_0(u) + qN_1(u)),}$$

where $\widehat{\delta}^$ denotes iteration of normalized input macro-steps. Hence two words with the same Parikh vector lead from the same normalized configuration to exactly the same normalized configuration.*

Proof. Injectivity was noted above. Every integer residual is reachable from zero: Bézout gives an integer solution of $qy - px = r$, and adding sufficiently many copies of (p, q) to (y, x) makes both counts nonnegative.

Suppose first that $r = pk + a_s > 0$. Reading 0 removes one unary 0, and

$$pk + a_s - p = p(k - 1) + a_s.$$

If $k > 1$ this is already the stable encoding of $r - p$; if $k = 1$, exposing Z_0 triggers the unique boundary normalization to $\eta(a_s) = \eta(r - p)$ (with $s = 0$ already equal to the zero encoding). For input 1, set

$$s' = (s + q) \bmod p, \quad c_s = \frac{a_s + q - a_{s'}}{p}.$$

The numerator is a multiple of p and is greater than $-p$, hence $c_s \geq 0$. The positive stable rule changes the unary length from k to $k + c_s$, and

$$pk + a_s + q = p(k + c_s) + a_{s'},$$

so the normalized target is $\eta(r + q)$.

Now let $r = -qk + b_t < 0$. Reading 1 removes one unary 1, and

$$-qk + b_t + q = -q(k - 1) + b_t.$$

At $k = 1$, exposing X triggers the unique boundary normalization to $\eta(b_t) = \eta(r + q)$. For input 0, put

$$t' = (t + p) \bmod q, \quad d_t = \frac{p - b_t + b_{t'}}{q}.$$

Again $d_t \geq 0$, and the negative stable rule uses

$$-qk + b_t - p = -q(k + d_t) + b_{t'},$$

which is exactly the encoding of $r - p$. At $r = 0$, the two binary rules are defined directly to $\eta(q)$ and $\eta(-p)$.

Thus every labeled residual edge has exactly the corresponding normalized configuration edge. The push-at-most-two compiler only replaces a semantic homogeneous write by a unique forced ε -suffix chain with the same stable target, so the macro-step identity is unchanged. Induction on $|u|$ gives the word formula. Since every residual occurs and η is injective, the labeled graphs are isomorphic. The Parikh statement follows immediately from the displayed formula. $\square$

Corollary 7.8 (Exact configuration-level rectangle lift). *Let $\mu(r) = (x_r, y_r)$. The map*

$$\widehat{\Psi}_r(i, j) = \eta(r - pi + qj), \quad 0 \leq i \leq x_r, \quad 0 \leq j \leq y_r,$$

is a label-preserving isomorphism from the shortest-completion rectangle $[0, x_r] \times [0, y_r]$ onto the normalized shortest-path interval from $\eta(r)$ to $\eta(0)$. In particular, every elementary square is a genuine normalized DPDA commuting diamond:

$$\boxed{\widehat{\delta}^*(\widehat{\Psi}_r(i, j), 01) = \widehat{\Psi}_r(i + 1, j + 1) = \widehat{\delta}^*(\widehat{\Psi}_r(i, j), 10)}$$

whenever $i < x_r$ and $j < y_r$.

Proof. Combine the graph isomorphism of theorem 7.7 with the residual rectangle theorem 7.3. The two length-two labels commute because both change the residual by $-p + q$ and normalization returns the unique stable encoding of that residual. $\square$

The distinction between the two configuration results is worth emphasizing. In the motivating 2:1 DPDA, theorem 5.1 is realized directly by ordinary solid input transitions. For the general compact DPDA, the exact rectangle is realized in the normalized stable configuration graph: one labeled macro-edge may contain a uniquely forced ε -normalization segment, but no additional input or nondeterministic choice.

Example: $p = 3, q = 2, r = 4$. Here $\mu(4) = (2, 1)$, so the abstract interval is a 3×2 rectangle. The stable encodings are

$$\begin{aligned} \eta(4) &= (R_1, 00Z_0), & \eta(1) &= (R_1, 0Z_0), & \eta(-2) &= (R_0, 1X), \\ \eta(6) &= (R_0, 00Z_0), & \eta(3) &= (R_0, 0Z_0), & \eta(0) &= (R_0, Z_0). \end{aligned}$$

Thus even the positive-to-negative boundary edge remains inside the exact normalized rectangle.

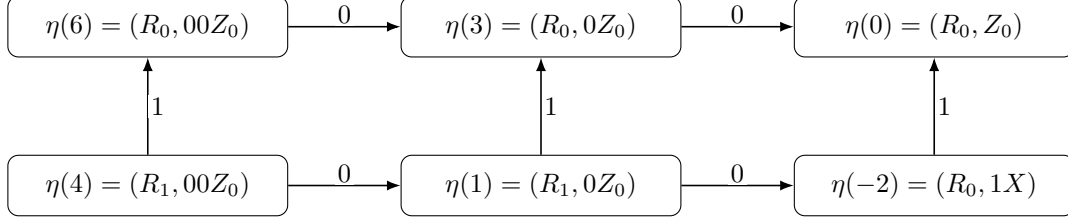


Figure 4: Exact normalized configuration rectangle for $p = 3, q = 2, r = 4$. Each displayed edge is one binary input symbol followed, when necessary, by its uniquely forced ε -normalization. The lower-right 0-edge crosses from positive residual 1 to negative residual -2 , while the rectangle and both commuting diamonds remain exact.

8 Eliminating normalization by changing coordinates

The four-symbol construction of section 7.1 realizes the residual graph exactly after deterministic normalization. The motivating 2:1 ladder, however, is stronger: its relevant edges are ordinary input transitions with no intervening ε -step. We first identify the exact obstruction in the companion four-symbol coordinates. We then separate two resource questions: with only one work symbol and a permanent bottom marker, raw conjugacy has exact state cost $p + q$; with two work symbols, a different coordinate choice realizes the complete residual graph using only $\max(p, q)$ states and three total stack symbols.

8.1 The exact defect set of the four-symbol homogeneous-tail encoding

Call a labeled residual edge (r, a) a *raw defect* of the encoding η if the immediate binary-input transition from $\eta(r)$ does not already equal $\eta(T_a(r))$, where

$$T_0(r) = r - p, \quad T_1(r) = r + q.$$

Thus a defect is precisely an edge on which the macro-step of theorem 7.7 needs a nonempty forced normalization segment.

Proposition 8.1 (Exact four-symbol raw-defect set). *For the right-endmarker four-symbol paired-sign encoding of section 7.1,*

$$\mathcal{D}_{p,q} = \{(s, 0) : 1 \leq s < p\} \cup \{(-j, 1) : 1 \leq j \leq q\}.$$

Hence

$$|\mathcal{D}_{p,q}| = p + q - 1.$$

Every labeled residual edge outside $\mathcal{D}_{p,q}$ is realized by one ordinary binary-input transition between stable configurations.

Proof. For $r = pk + a_s > 0$, input 1 always uses the positive stable rule and lands directly in $\eta(r + q)$. Input 0 merely pops one unary 0. If $k > 1$, the result is the stable encoding $\eta(r - p)$. When $k = 1$, one has $r = a_s + p$: for $s = 0$ this is $r = p$ and the pop already leaves the zero encoding (R_0, Z_0) , whereas for $1 \leq s < p$ one has $r = s$ and the pop leaves the transient pair (R_s, Z_0) , which requires the boundary normalization to $\eta(s - p)$. These are exactly the positive defects $(s, 0)$.

For $r = -qk + b_t < 0$, input 0 always lands directly in the next negative stable encoding. Input 1 pops one unary 1. If $k > 1$ the result is stable. If $k = 1$, the exposed sentinel is X and a boundary normalization is required for every t : the corresponding residual is $-q$ when $t = 0$ and $-t$ when $1 \leq t < q$. Thus the negative defects are precisely $(-j, 1)$ for $1 \leq j \leq q$. Zero itself has direct binary rules in both directions. The two defect families are disjoint and have sizes $p - 1$ and q . $\square$

Remark 8.2 (Boundary scale). For every nontrivial coprime pair, the cardinality $|\mathcal{D}_{p,q}| = p + q - 1$ coincides numerically with the completion-frontier parameter $K_{p,q} = p + q - 1$ of the companion analysis [11]. The two quantities answer different questions—one counts raw normalization defects in this four-symbol encoding, the other counts the maximal finite completion frontier—so the equality is recorded as a structural coincidence of the same boundary scale rather than as an identification of the notions. The degenerate pair $(1, 1)$ is exceptional: the generic four-symbol orientation has one raw defect, while the companion frontier is zero.

These defects are not an accident of the particular transition table once the physical encoding η has been fixed.

Proposition 8.3 (Boundary obstruction for the fixed homogeneous-tail encoding). *Keep the four-symbol physical representations $\eta(r)$ of section 7.1. No standard PDA transition function can realize every edge in $\mathcal{D}_{p,q}$ as a single input transition while simultaneously preserving the required one-step behavior on the adjacent length-two unary tails. In particular, all $p + q - 1$ defects of Proposition 8.1 are unavoidable within this fixed encoding.*

Proof. Fix $1 \leq s < p$. The configurations

$$\eta(p + s) = (R_s, 00Z_0), \quad \eta(s) = (R_s, 0Z_0)$$

have the same state and top symbol. Correct one-step realization of $p + s \xrightarrow{0} s$ forces the rule on $(R_s, 0)$ to stay in R_s and pop the top 0, because the exact target is $(R_s, 0Z_0)$. The same rule applied to $(R_s, 0Z_0)$ therefore yields (R_s, Z_0) , not the required negative encoding $\eta(s - p)$. Hence $(s, 0)$ cannot simultaneously be raw.

For $0 \leq t < q$, compare

$$\eta(-2q + b_t) = (R_t, 11X), \quad \eta(-q + b_t) = (R_t, 1X).$$

Correct one-step realization of the first input-1 edge forces the rule on $(R_t, 1)$ to stay in R_t and pop one 1. Applying that same rule to the length-one tail leaves (R_t, X) , whereas the correct target after the next 1 is $\eta(b_t)$, which is zero for $t = 0$ and positive for $t > 0$. Thus every negative boundary edge is likewise unavoidable. The argument uses only the fixed state/top representations, not the particular normalization rules chosen afterward. $\square$

Corollary 8.4 (Raw rectangle criterion for the four-symbol orientation). *For the four-symbol right-endmarker encoding, the complete shortest-completion rectangle from residual r is realized entirely by raw one-input stable-to-stable edges if and only if*

$$\boxed{r = 0 \quad \text{or} \quad (r > 0 \text{ and } p \mid r).}$$

Proof. If $r > 0$ and $p \mid r$, then $y_r = 0$ and the rectangle is the single zero-labeled line through positive multiples of p down to zero; none of its edges belongs to Proposition 8.1. If $r > 0$ and $p \nmid r$, write $s = r \bmod p \in \{1, \dots, p-1\}$. Since $y_r > 0$, the zero-first boundary path eventually contains the defect $s \xrightarrow{0} s - p$. If $r < 0$, the rightmost column of the rectangle contains the final edge $-q \xrightarrow{1} 0$, which is always a defect. The zero rectangle is trivial. $\square$

The asymmetry in Corollary 8.4 belongs to the chosen representation: zero shares the positive sentinel Z_0 , whereas the negative side uses X . The next construction removes this orientation artifact entirely.

8.2 Exact one-work-symbol baseline

The preceding defect analysis suggests a sharper resource question. Suppose the stack has only one work symbol in addition to a permanent bottom marker,

$$\Gamma_2 = \{A, Z_0\}.$$

We keep the semantic requirement fixed: an injective residual encoding must realize each binary residual update in one input transition, with no ε -normalization. The marker Z_0 occurs exactly once at the bottom and is never inserted above an existing stack suffix, so every encoded stack has the form $A^h Z_0$. Replacement words may have arbitrary fixed finite length, as in the raw constructions above.

Theorem 8.5 (Exact one-work-symbol raw-conjugacy complexity). *Fix coprime $p, q \geq 1$. In the permanent-bottom, one-work-symbol model $\Gamma_2 = \{A, Z_0\}$, every ε -free raw residual-configuration conjugacy for $\mathcal{R}_{p,q}$ requires*

$$|Q| \geq p + q.$$

Conversely, there is such a right-endmarker DPDA with exactly

$$|Q| = p + q, \quad |\Gamma| = 2.$$

Thus $p + q$ is the exact state complexity of raw residual conjugacy in this one-work-symbol model.

Proof. Let $\eta(r) = (q_r, A^{h_r} Z_0)$ be any injective raw encoding, where $h_r \geq 0$. On configurations with top A , an input transition replaces one visible A by a fixed word A^m and therefore changes the height by $m - 1 \geq -1$.

Consider first input 0. The map $F_0 : \eta(r) \mapsto \eta(r - p)$ has exactly p bi-infinite orbits, one for each residual class modulo p . Along any such orbit, the bottom marker can be exposed only finitely many times: there are only $|Q|$ bottom configurations (q, Z_0) , and a repeated one would contradict injectivity of η . Hence, on the $r \rightarrow +\infty$ tail, the state sequence eventually runs around a directed cycle of the top- A state map for input 0.

Let that cycle have length L and net height change D in one traversal. Moving forward by F_0 decreases the residual. On the left-infinite $r \rightarrow +\infty$ tail, nonnegativity of the stack height rules out $D > 0$, while $D = 0$ would repeat an exact configuration. Therefore $D = -d < 0$. At a fixed phase state on the cycle, one orbit occupies one residue class of heights modulo d ; two different F_0 -orbits with the same class would eventually meet in the same state and height and hence be the same orbit. Thus the cycle carries at most d residual classes. Since every single transition decreases height by at most one, $d \leq L$. Summing over the disjoint negative-drift F_0 cycles that serve the p positive-end residue classes gives a set P of states with

$$|P| \geq p.$$

Apply the symmetric argument to input 1. The map $F_1 : \eta(r) \mapsto \eta(r + q)$ has q bi-infinite orbits. Their $r \rightarrow -\infty$ tails require negative-drift top- A cycles for F_1 , and the same height-residue argument gives a set N of states with

$$|N| \geq q.$$

The sets P and N are disjoint. Indeed, suppose a state lay on both a positive-end negative-drift F_0 cycle and a negative-end negative-drift F_1 cycle. The first role supplies configurations in that state with arbitrarily large positive residual and arbitrarily large height. Starting from any such configuration and repeatedly traversing the negative-drift F_1 cycle must eventually expose Z_0 , while the residual only increases. Choosing larger and larger starting residuals would therefore produce infinitely many distinct positive residuals encoded at one of the finitely many bottom configurations, contradicting injectivity. Hence

$$|Q| \geq |P| + |N| \geq p + q.$$

For tightness, use disjoint state families

$$P_0, \dots, P_{p-1}, \quad N_0, \dots, N_{q-1}.$$

For $r \geq 0$, write $r = pk + s$ with $k \geq 0$ and $0 \leq s < p$ and set

$$\xi(r) = (P_s, A^k Z_0).$$

For $r < 0$, write uniquely $r = u - qh$ with $0 \leq u < q$ and $h \geq 1$, and set

$$\xi(r) = (N_u, A^{h-1} Z_0).$$

For $0 \leq s < p$ define

$$s^+ = (s + q) \bmod p, \quad c_s = \frac{s + q - s^+}{p} \geq 0,$$

and write

$$s - p = u_s - qh_s, \quad 0 \leq u_s < q, \quad h_s \geq 1.$$

Use

$$\begin{aligned} \delta(P_s, 0, A) &= (P_s, \varepsilon), & \delta(P_s, 1, A) &= (P_{s^+}, A^{c_s+1}), \\ \delta(P_s, 1, Z_0) &= (P_{s^+}, A^{c_s} Z_0), & \delta(P_s, 0, Z_0) &= (N_{u_s}, A^{h_s-1} Z_0). \end{aligned}$$

For $0 \leq u < q$, put

$$u^- = (u - p) \bmod q, \quad d_u = \frac{u^- - u + p}{q} \geq 0.$$

Then use

$$\begin{aligned} \delta(N_u, 1, A) &= (N_u, \varepsilon), & \delta(N_u, 0, A) &= (N_{u^-}, A^{d_u+1}), \\ \delta(N_u, 1, Z_0) &= \xi(u), & \delta(N_u, 0, Z_0) &= (N_{u^-}, A^{d_u} Z_0). \end{aligned}$$

The identities defining c_s, h_s, d_u verify directly that these rules send $\xi(r)$ to $\xi(r + q)$ on input 1 and to $\xi(r - p)$ on input 0. The right endmarker is accepted only at $\xi(0) = (P_0, Z_0)$. No ε -transition is used. $\square$

Corollary 8.6 (Alphabet minimality at the $\max(p, q)$ state budget). *Within the same permanent-bottom raw-conjugacy model, no DPDA with*

$$|Q| \leq \max(p, q)$$

can use only one work symbol. Therefore the three-symbol construction of theorem 8.7 uses the minimum possible total stack-alphabet size at that state budget:

$$\boxed{|Q| = \max(p, q), \quad |\Gamma|_{\min} = 3.}$$

This is a lower bound for exact raw residual conjugacy, not for unrestricted DPDA recognition of $L_{p,q}$.

8.3 Boundary-compatible three-symbol encoding

The preceding obstruction is tied to the shifted homogeneous-tail coordinates of section 7.1. It does *not* imply that a fifth symbol, or even the fourth symbol X , is necessary. A different alignment of the finite residual bases makes the sign change coincide with an ordinary unary pop.

Assume first that $p \geq q$. Let

$$Q_3 = \{R_0, \dots, R_{p-1}\}, \quad \Gamma_3 = \{0, 1, Z_0\}.$$

For $r \geq 0$, write uniquely

$$r = pk + s, \quad k \geq 0, \quad 0 \leq s < p,$$

and define

$$\boxed{\zeta(r) = (R_s, 0^k Z_0)}.$$

For $r < 0$, write uniquely

$$r = u - qh, \quad h \geq 1, \quad 0 \leq u < q,$$

and define

$$\boxed{\zeta(r) = (R_u, 1^h Z_0)}.$$

Thus the states $R_0, \dots, R_{q-1}$ are deliberately shared by a nonnegative boundary role and a negative phase. The sign remains visible from the top symbol: a negative configuration always has top 1, whereas a nonnegative configuration has top 0 or Z_0 .

For $0 \leq s < p$, put

$$s^+ = (s + q) \bmod p, \quad c_s = \frac{s + q - s^+}{p} \in \{0, 1\},$$

and

$$u_s = (s - p) \bmod q, \quad h_s = \frac{u_s + p - s}{q} \geq 1.$$

For $0 \leq u < q$, put

$$u^- = (u - p) \bmod q, \quad d_u = \frac{u^- - u + p}{q} \geq 1.$$

Use the positive-tail rules

$$\boxed{\delta(R_s, 0, 0) = (R_s, \varepsilon), \quad \delta(R_s, 1, 0) = (R_{s^+}, 0^{c_s+1}),}$$

the nonnegative-boundary rules

$$\boxed{\delta(R_s, 1, Z_0) = (R_{s^+}, 0^{c_s} Z_0), \quad \delta(R_s, 0, Z_0) = (R_{u_s}, 1^{h_s} Z_0),}$$

and, for $0 \leq u < q$, the negative rules

$$\boxed{\delta(R_u, 1, 1) = (R_u, \varepsilon), \quad \delta(R_u, 0, 1) = (R_{u^-}, 1^{d_u+1}).}$$

The mandatory right endmarker is accepted only at

$$\zeta(0) = (R_0, Z_0).$$

There are no ε -transitions. When $q > p$, apply the symmetric construction obtained by exchanging 0 and 1, exchanging p and q , and replacing residual r by $-r$.

Theorem 8.7 (Three-symbol raw residual–configuration conjugacy). *For every coprime $p, q \geq 1$, there is an ε -free right-endmarker DPDA with*

$$\boxed{|Q| = \max(p, q), \quad |\Gamma| = 3}$$

whose stack alphabet is $\{0, 1, Z_0\}$ and whose stable encoding $\zeta : \mathbb{Z} \rightarrow Q \times \Gamma^$ satisfies, for every residual r ,*

$$\boxed{\delta(\zeta(r), 0) = \zeta(r - p), \quad \delta(\zeta(r), 1) = \zeta(r + q).}$$

The right endmarker is accepted exactly at $\zeta(0)$, so the machine recognizes $L_{p,q}$. Consequently ζ is a label-preserving graph isomorphism from $\mathcal{R}_{p,q}$ onto the reachable binary-input configuration graph, and for every $w \in \{0, 1\}^$,*

$$\boxed{\delta^*(\zeta(r), w) = \zeta(r - pN_0(w) + qN_1(w)).}$$

In particular, ordinary runs from the same configuration synchronize exactly on consumed Parikh vectors, with no normalization quotient or macro-step convention.

Proof. Assume $p \geq q$; the other case follows by the stated involution. The encoding is injective. Nonnegative residuals have the unique Euclidean form $pk + s$. Negative residuals have the unique form $u - qh$ with $0 \leq u < q$ and $h \geq 1$. The two sign families are disjoint because the negative family has top 1, while the nonnegative family has top 0 or Z_0 .

Let $r = pk + s \geq 0$. If $k \geq 1$, reading 0 pops one 0 and gives

$$pk + s - p = p(k - 1) + s,$$

which is exactly $\zeta(r - p)$. Reading 1 changes the residue class from s to s^+ and uses

$$pk + s + q = p(k + c_s) + s^+,$$

so replacing the visible 0 by 0^{c_s+1} gives $\zeta(r + q)$.

If $k = 0$, the top is Z_0 . The input-1 rule uses the same carry identity with $k = 0$. For input 0,

$$s - p = u_s - qh_s,$$

so the boundary rule writes exactly the negative encoding $\zeta(s - p)$ in the same input step. This includes $s = 0$, hence the zero-to-negative move.

Now let $r = u - qh < 0$. Reading 1 pops one 1. If $h > 1$, the result is the negative encoding $u - q(h - 1)$. If $h = 1$, the pop exposes Z_0 and leaves (R_u, Z_0) , which is precisely the nonnegative boundary encoding $\zeta(u)$. Thus the same raw rule handles both the interior and the sign-changing boundary.

For input 0, the identity

$$u - p = u^- - qd_u$$

gives

$$r - p = u^- - q(h + d_u).$$

Replacing one visible 1 by 1^{d_u+1} therefore produces exactly $\zeta(r - p)$. No case uses an ε -move.

The right endmarker is enabled only at $(R_0, Z_0) = \zeta(0)$, so acceptance is equivalent to zero residual. The one-step identities imply the word formula by induction. Residual surjectivity and injectivity of ζ give the labeled graph isomorphism.

All replacement words are finite constants determined by the fixed pair (p, q) . As with the earlier semantic constructions, no push-at-most-two restriction is imposed here; compiling a long fixed write into bounded writes would introduce auxiliary steps and would no longer be a raw one-transition realization. $\square$

Theorem 8.8 (Exact state complexity in the sign-separated homogeneous-tail model). *Fix the permanent-bottom alphabet*

$$\Gamma_3 = \{0, 1, Z_0\},$$

and consider ε -free raw residual-configuration encodings in which every sufficiently positive residual has stack form $0^h Z_0$ and every sufficiently negative residual has stack form $1^h Z_0$. Then every such encoding requires

$$|Q| \geq \max(p, q).$$

The construction of theorem 8.7 attains equality. Hence

$$|Q|_{\min} = \max(p, q)$$

inside this sign-separated homogeneous-tail raw-conjugacy model.

Proof. Consider the input-0 action on the positive end. Its residual map

$$r \mapsto r - p$$

has exactly p bi-infinite residue classes modulo p . Because there are only finitely many bottom configurations (q, Z_0) and the residual encoding is injective, each such class is eventually represented with nonempty stack $0^h Z_0$ as $r \rightarrow +\infty$.

On a sufficiently deep positive configuration, a raw input-0 transition must map a homogeneous zero tail to another homogeneous zero tail. Thus, for every state that occurs arbitrarily far on the positive end, the rule on top 0 has the form

$$\delta(q, 0, 0) = (q', 0^m)$$

for a fixed $m \geq 0$. It therefore induces a finite directed state map, and the height change of one step is $m - 1 \geq -1$.

Fix one directed cycle of length L in this positive-end state map and let its total height change in one forward traversal be D . Along the residual direction $r \mapsto r - p$, one cannot have $D > 0$: tracing the same cycle backward toward arbitrarily large positive residuals would eventually force negative stack height. One cannot have $D = 0$ either, because the same state and height would then repeat after one cycle, contradicting injectivity. Hence

$$D = -d$$

for some integer $d \geq 1$. Since each of the L transitions decreases height by at most one,

$$d \leq L.$$

At any fixed phase state of this cycle, one residual class modulo p occupies one congruence class of stack heights modulo d . Two distinct residual classes using the same phase and the same height congruence would eventually meet in the same state and at the same height, contradicting injectivity. Thus this cycle can carry at most $d \leq L$ of the p positive-end residue classes. Summing over the disjoint positive-end cycles shows that at least p control states are required:

$$|Q| \geq p.$$

Apply the symmetric argument to the input-1 action on the negative end. There are exactly q residual classes modulo q , sufficiently negative configurations have homogeneous stacks $1^h Z_0$, and every deep input-1 rule changes height by at least -1 . The same cycle-and-height-residue argument yields

$$|Q| \geq q.$$

Therefore

$$|Q| \geq \max(p, q).$$

The three-symbol construction of theorem 8.7 uses exactly $\max(p, q)$ states and belongs to the stated sign-separated homogeneous-tail class, so the lower bound is attained. $\square$

Corollary 8.9 (Exact homogeneous-tail alphabet frontier). *Consider permanent-bottom, ε -free raw residual-configuration conjugacies with an arbitrary finite work alphabet of size $k \geq 1$. Assume only that every positive residue class modulo p is eventually represented by an unbounded homogeneous tail $X_s^h Z_0$, and every negative residue class modulo q is eventually represented by an unbounded homogeneous tail $Y_t^h Z_0$; the symbols X_s and Y_t may a priori depend on the residue class. Let $S_{\text{hom}}(k)$ be the minimum number of control states in this model. Then*

$$S_{\text{hom}}(k) = \begin{cases} p + q, & k = 1, \\ \max(p, q), & k \geq 2. \end{cases}$$

Thus, within the asymptotically homogeneous-tail raw-conjugacy model, the second work symbol gives the entire possible state reduction; additional work symbols do not reduce the exact state minimum further.

Proof. First consider the positive end. Let a sufficiently large positive residual in class $s \pmod{p}$ have homogeneous tail $X_s^h Z_0$, with h arbitrarily large. Reading 1 sends its residual class to

$$s^+ = (s + q) \pmod{p}.$$

For sufficiently deep source and target configurations, the one-step PDA rewrite changes only the visible top symbol and preserves the suffix $X_s^{h-1} Z_0$. Since the target is again a homogeneous tail in class s^+ , this forces $X_s = X_{s^+}$. Addition by q is a single cycle on the p residue classes, so all positive-end tail symbols coincide; call the common symbol X_+ .

Symmetrically, reading 0 permutes the negative residue classes modulo q by subtraction of p . Coprimality makes this a single cycle, and the same preserved-suffix argument forces all negative-end homogeneous tails to use one common symbol X_- . The cycle-and-height argument of theorem 8.8 therefore still forces at least p positive-end state roles and at least q negative-end state roles.

If $k = 1$, necessarily $X_+ = X_-$. The disjointness argument from theorem 8.5 applies to the two deep homogeneous ends: a state serving both a positive-end negative-drift input-0 cycle and a negative-end negative-drift input-1 cycle would drive arbitrarily many distinct residuals to finitely many bottom configurations. Hence the two state families are disjoint and $|Q| \geq p + q$. Equality is attained by the one-work-symbol construction of theorem 8.5.

If $k \geq 2$, the two end symbols may be chosen distinct, so the lower bounds $|Q| \geq p$ and $|Q| \geq q$ give $|Q| \geq \max(p, q)$. The boundary-compatible construction of theorem 8.7 uses two work symbols and exactly $\max(p, q)$ states, and therefore attains the bound for every $k \geq 2$ as well. $\square$

Theorem 8.10 (Exact three-symbol state complexity for the motivating ratio). *For $(p, q) = (2, 1)$, consider the full permanent-bottom three-symbol model*

$$\Gamma_3 = \{A, B, Z_0\}$$

with no ε -moves and an injective raw residual-configuration conjugacy, but impose no homogeneous-tail or sign-separated form on the encoding. Then

$$|Q| \geq 2.$$

The construction of theorem 8.7 has two states for $(2, 1)$, so the bound is tight. Hence

$$\boxed{|Q|_{\min} = 2}$$

for three-symbol raw conjugacy of the motivating ratio in the permanent-bottom model.

Proof. Assume for contradiction that there is only one control state. Write S_r for the physical stack encoding residual r , and let

$$G : S_r \mapsto S_{r+1}, \quad F : S_r \mapsto S_{r-2}$$

be the raw input-1 and input-0 actions. On the encoded orbit,

$$FG^2 = G^2F = \text{id}.$$

Count only work symbols above the permanent bottom marker. Choose a residual r_* whose encoded stack has minimum work height and put $T_k = S_{r_*+k}$. Then

$$G(T_k) = T_{k+1}, \quad F(T_k) = T_{k-2}.$$

First note a uniform replacement bound. If a work-top rule of G replaces one symbol by a word u , then after the first G step the work height changes by $|u| - 1$, while the second G step can delete at most one current work symbol. The following F step returns to the original configuration and can again delete at most one current work symbol. Since $FG^2 = \text{id}$,

$$|u| \leq 3.$$

If the minimum configuration is the bare bottom stack and G writes uZ_0 from Z_0 , the same argument has only $|u|$ newly written work symbols and gives

$$|u| \leq 2.$$

Suppose first that the minimum work height is positive. Write

$$T_0 = X\beta,$$

where $X \in \{A, B\}$ and β is the suffix below the top symbol. Every residual is reachable from r_* by a binary word, and no encoded stack is shorter than T_0 . Therefore every T_k retains β as a suffix: a transition may rewrite the current top, but it can never expose β as an encoded stack because that would have smaller work height. Strip this common inert suffix and denote it by $\#$; after exchanging A and B if necessary, the reduced minimum stack is

$$T_0 = A\#,$$

and every reduced encoded stack contains at least one work symbol above $\#$.

Because $F(T_2) = T_0$ and a one-step PDA rewrite preserves the suffix below the source top, T_2 can only be

$$B\#, \quad AA\#, \quad BA\#.$$

Solving $T_1 = G(T_0)$ and $T_2 = G(T_1)$ with the bound $|G(A)|, |G(B)| \leq 3$ first gives eight algebraic rule pairs. Two of them, $(G(A), G(B)) = (B, B)$ and (BA, B) , already satisfy $T_1 = T_2$ and are

excluded immediately by injectivity. The remaining six pairs are exactly the rows below. The last column gives the contradiction forced by $F(T_3) = T_1$, minimality, or injectivity.

$G(A)$	$G(B)$	T_1	T_2	T_3	obstruction
B	AA	$B\#$	$AA\#$	$BA\#$	$\text{tail}(T_3) = A\# \not\leq_{\text{suf}} B\#$
B	BA	$B\#$	$BA\#$	$BAA\#$	$F(B) = \varepsilon \Rightarrow F(T_3) = AA\#$
BA	A	$BA\#$	$AA\#$	$BAA\#$	$AA\# \not\leq_{\text{suf}} BA\#$
BB	ε	$BB\#$	$B\#$	$\#$	smaller than the minimum
BAA	ε	$BAA\#$	$AA\#$	$BAAA\#$	$AAA\# \not\leq_{\text{suf}} BAA\#$
BBA	ε	$BBA\#$	$BA\#$	$A\#$	$T_3 = T_0$.

Here $\leq_{\text{suf}}$ means “is a suffix of.” The table is complete because the target T_2 has work length at most two; if its length is two its lower work symbol must be A , while length one cannot give $A\#$ again by injectivity. Substituting the three possible targets into $G(T_1) = T_2$ gives the eight algebraic pairs described above; the two immediate collisions leave precisely the displayed six.

It remains to consider minimum work height zero. Then $T_0 = Z_0$. The bottom-write bound gives

$$G(Z_0) = wZ_0, \quad 1 \leq |w| \leq 2.$$

Rename the first work symbol of w as A . Up to exchanging A and B , there are only three possibilities: $w = A, AA, AB$.

If $w = AA$, the condition $F(T_2) = T_0$ forces $G(A) = \varepsilon$, so $T_2 = AZ_0$ and $T_3 = Z_0 = T_0$, contradicting injectivity.

If $w = A$, then $F(T_2) = T_0$ forces $G(A) = B$. Next $F(T_3) = T_1$ leaves only

$$G(B) = AA \quad \text{or} \quad G(B) = BA.$$

In the first case $T_4 = BAZ_0$ and in the second $T_4 = BAAZ_0$. Since $F(B) = \varepsilon$ is already forced by $F(T_2) = T_0$, neither stack can map by F to $T_2 = BZ_0$.

Finally let $w = AB$. Then $F(T_2) = T_0$ forces $G(A) = \varepsilon$ and $T_2 = BZ_0$, hence also $F(B) = \varepsilon$. The condition $F(T_3) = T_1$ leaves exactly

$$G(B) \in \{A, AAB, BAB\}.$$

These give respectively $T_4 = T_0$, $T_4 = T_1$, or $T_4 = BABABZ_0$; in the last case the already forced pop $F(B) = \varepsilon$ cannot send T_4 to $T_2 = BZ_0$. Each possibility is impossible, and the symbol-exchanged cases are identical. Thus one state cannot realize the raw residual graph. $\square$

This theorem removes the homogeneous-tail hypothesis from the state lower bound at the motivating ratio itself. For general coprime (p, q) , whether an arbitrary three-symbol raw-conjugate encoding can beat $\max(p, q)$ states remains open.

Theorem 8.11 (Exact three-symbol state complexity for the 3:1 ratio). *For $(p, q) = (3, 1)$, consider the full permanent-bottom three-symbol model*

$$\Gamma_3 = \{A, B, Z_0\}$$

with no ε -moves and an injective raw residual-configuration conjugacy, with no homogeneous-tail or sign-separated assumption on the encoding. Then

$$\boxed{|Q| \geq 3.}$$

The construction of theorem 8.7 has three states for $(3, 1)$, so the bound is tight:

$$\boxed{|Q|_{\min} = 3.}$$

By exchanging the two input symbols and reversing the residual sign, the same exact bound holds for $(p, q) = (1, 3)$.

Proof. Assume for contradiction that two control states suffice. Write T_k for the configuration encoding residual $r_* + k$, where r_* is chosen so that T_0 has minimum work-stack height. Let

$$G : T_k \mapsto T_{k+1}, \quad F : T_k \mapsto T_{k-3}$$

be the raw input-1 and input-0 actions. On the encoded orbit,

$$\boxed{FG^3 = G^3F = \text{id.}}$$

As in theorem 8.10, this identity gives a finite replacement bound. If a work-top rule of G writes a word of length m , then after that first step the two remaining G -moves and the final F -move can delete at most three work symbols before the original configuration must be restored. Hence $m - 1 \leq 3$, so every work-top replacement of G has length at most four. Applying the same argument to $G^3F = \text{id}$ gives the same bound for work-top rules of F . From a bare-bottom configuration, the first step creates m work symbols rather than replacing one, so every bottom write has length at most three.

We use only two elementary PDA facts in the finite reduction below. First, a top rewrite preserves the entire suffix below the source top. Second, a repeated state/top pair must use the same transition rule. Thus, once T_{k+3} is known, the requirement

$$F(T_{k+3}) = T_k$$

either determines the corresponding F -rule uniquely or is impossible because the preserved source tail is not a suffix of the required target. Likewise, once a G -rule has appeared, all later uses of the same state/top pair are forced. These observations make the following case reduction finite and exhaustive.

Suppose first that the minimum work height is positive. Write

$$T_0 = (q_0, X\beta).$$

The suffix β can never be exposed by an encoded configuration, since that would have smaller work height. It is therefore an inert common suffix along the orbit and may be stripped. Rename the two states and the two work symbols so that the reduced minimum configuration is

$$T_0 = (0, A\#),$$

where $\#$ denotes the fixed inert suffix and every reduced encoded stack contains at least one work symbol above it.

Starting from T_0 , repeatedly impose

$$G(T_k) = T_{k+1}, \quad F(T_{k+3}) = T_k,$$

the length-four replacement bound, nonemptiness above $\#$, injectivity, suffix preservation, and deterministic reuse of every state/top rule. Through T_7 , all branches reduce to exactly the following four prefixes:

case	$(q_0, q_1, \dots, q_7)$	$(\text{stk } T_0, \dots, \text{stk } T_7)$
I _a	00101101	$A, BBBA, BBA, BA, A, BBBA, BBA, BA$
I _b	01111000	$A, BBBA, BBA, BA, A, BBBA, BBA, BA$
II _a	01011010	$A, BBAA, BAA, AA, BBBAA, BBAA, BAA, AA$
II _b	01110000	$A, BBAA, BAA, AA, BBBAA, BBAA, BAA, AA$

Every displayed stack implicitly ends in $\#$. The table is a finite reduction certificate: at a first occurrence of a state/top pair there are only the bounded replacement words to consider;

at every later occurrence the G -rule is fixed, while the relation $F(T_{k+3}) = T_k$ either fixes the corresponding F -rule or rejects the branch by suffix incompatibility. No further transition freedom remains in any of the four rows.

Cases I_a and I_b both force

$$T_8 = T_0,$$

contradicting injectivity. In cases II_a and II_b , the already fixed rule on the source cell of T_7 forces

$$T_8 = (1, BBAAA\#).$$

But $F(T_8)$ must equal $T_5 = (0, BBAAA\#)$. Any one-step rewrite of T_8 preserves the tail $BAAA\#$ below its top symbol, and $BAAA\#$ is not a suffix of $BBAAA\#$. Hence $F(T_8) = T_5$ is impossible.

It remains to consider minimum work height zero. Normalize

$$T_0 = (0, \#),$$

where now $\#$ is the bare bottom marker. Apply the same finite reduction, allowing bottom writes of length at most three. Up to exchanging A and B , the only prefix consistent through T_{10} is

k	0	1	2	3	4	5	6	7	8	9	10
q_k	0	1	1	1	0	0	0	0	1	1	1
$\text{stk } T_k$	$\#$	$AAB\#$	$AB\#$	$B\#$	$AAAB\#$	$AAB\#$	$AB\#$	$B\#$	$A\#$	$\#$	$BB\#$.

Here the occurrence $T_3 = (1, B\#) \mapsto T_4 = (0, AAAB\#)$ fixes the G -rule on $(1, B)$, so applying that same rule to $T_{10} = (1, BB\#)$ forces

$$T_{11} = (0, AAABB\#).$$

Also $F(T_4) = T_1$ has already fixed the F -rule on $(0, A)$ as a pop into state 1. Consequently

$$F(T_{11}) = (1, AAABB\#) \neq T_8 = (1, A\#),$$

contradicting $F(T_{11}) = T_8$.

Both possible minimum-height cases are impossible. Thus two states cannot realize the raw residual graph, and the three-state construction is optimal in the stated model. $\square$

Theorem 8.12 (Computer-assisted exact three-symbol state complexity for the 3:2 ratio). *For $(p, q) = (3, 2)$, consider the full permanent-bottom three-symbol model*

$$\Gamma_3 = \{A, B, Z_0\}$$

with no ε -moves and an injective raw residual-configuration conjugacy, with no homogeneous-tail or sign-separated assumption on the encoding. Then

$$\boxed{|Q| \geq 3.}$$

The construction of theorem 8.7 uses three states for $(3, 2)$, so the bound is tight:

$$\boxed{|Q|_{\min} = 3.}$$

By exchanging the input symbols and reversing the residual sign, the same exact bound holds for $(p, q) = (2, 3)$.

Computer-assisted proof. Assume that two control states suffice. Let η be the injective raw encoding and choose an encoded configuration of minimum work-stack height. Write the two parity chains relative to its residual r_* as

$$A_k = \eta(r_* + 2k), \quad B_k = \eta(r_* + 1 + 2k).$$

Let G denote the raw input-1 action and F the raw input-0 action. Since G adds 2 to the residual and F subtracts 3,

$$G(A_k) = A_{k+1}, \quad G(B_k) = B_{k+1},$$

and

$$F(A_{k+2}) = B_k, \quad F(B_{k+1}) = A_k.$$

Equivalently, on the encoded residual graph,

$$\boxed{F^2 G^3 = G^3 F^2 = \text{id.}}$$

This identity makes the remaining search finite. A single consuming PDA transition can delete at most one work symbol. If a work-top rule writes m work symbols in place of one, its first step raises the work height by $m - 1$; the other four moves of either five-step inverse identity can remove at most four symbols before the original configuration must be recovered. Hence every work-top replacement of either F or G has length at most five. From a bare-bottom configuration the first step creates, rather than replaces, its work word, so every bottom write has length at most four.

As in theorem 8.11, if the minimum work height is positive, the suffix below the top of a minimum configuration can never be exposed by an encoded configuration and is therefore inert. Strip that common suffix and normalize the minimum to

$$A_0 = (0, A\#),$$

where $\#$ denotes the fixed inert suffix. If the minimum height is zero, normalize instead to

$$A_0 = (0, \#),$$

with $\#$ the bare bottom marker. Renaming the two states and, in the positive-minimum case, the visible work symbol loses no generality.

The finite certificate now enumerates all deterministic rules that can be encountered from these normalized minima under the replacement bounds above. A first occurrence of a state/top pair branches over every legal target state and bounded replacement word; a repeated pair reuses the already fixed rule. Whenever one of the displayed F -relations is due, the target is fixed, so suffix preservation either determines the unique possible replacement or rejects the branch. Injectivity rejects repeated encoded configurations.

The enumeration additionally uses only consequences of the same inverse identities as deletion-capacity pruning. For example,

$$F^2(A_3) = A_0, \quad F(B_1) = A_0,$$

force

$$|A_3| \leq |A_0| + 2, \quad |B_1| \leq |A_0| + 1,$$

where $|\cdot|$ denotes work height. More generally,

$$F(B_j) = A_{j-1}, \quad F(A_{j+2}) = B_j,$$

and, for the corresponding three-step G advances,

$$F^2(B_j) = B_{j-3}, \quad F^2(A_{j+2}) = A_{j-1},$$

give the analogous one- and two-deletion height bounds used by the certificate. These tests discard only configurations that could not return to the already prescribed target in the available number of one-pop transitions.

For positive minimum height, the exact survivor counts at the successive synthesis stages are

$$1, 123, 11643, 1691, 2120, 1591, 348, 43, 13, 0.$$

Thus no two-state candidate extends through the required next A -configuration. The search examines 1,130,430 branch attempts in this case. For minimum height zero, the survivor counts are

$$1, 61, 7556, 1134, 2146, 2966, 6846, 7032, 2376, 650, 190, 26, 0,$$

so again no candidate survives; this case examines 1,696,120 branch attempts. The two cases exhaust all possible minimum heights, and the bounded enumeration exhausts every rule choice allowed by the symbolic replacement bounds. Therefore a two-state raw conjugacy is impossible.

The complete certificate is implemented in the distributed file `second_paper_validation_suite.py`, using only the Python standard library. The script asserts the two survivor vectors above, the exact branch counts, and zero final survivors. The computation is used only for the finite bounded elimination after the mathematical reduction to a finite search space. $\square$

Theorems 8.10, 8.11, and 8.12 therefore show that the homogeneous-tail state lower bound survives arbitrary permanent-bottom three-symbol encodings for $(2, 1)$, $(3, 1)$, and the first genuinely two-sided case $(3, 2)$, together with their symmetric partners. The corresponding theorem for arbitrary coprime (p, q) remains open.

Corollary 8.13 (Exact cost of removing the right endmarker). *Under ordinary final-state acceptance on the binary alphabet, with no right endmarker, the exact state counts increase by one in the two raw-conjugacy models above:*

$$\boxed{|Q| = p + q + 1, \quad |\Gamma| = 2}$$

for the permanent-bottom one-work-symbol model, and

$$\boxed{|Q| = \max(p, q) + 1, \quad |\Gamma| = 3}$$

for the sign-separated homogeneous-tail model. Both bounds are tight.

Proof. For the one-work-symbol model, theorem 8.5 already forces $p + q$ states on the unbounded nonzero tails. Every one of those states occurs at some nonzero residual. Under ordinary final-state acceptance, a state used by any nonzero residual cannot be final, because ending the input in that configuration would otherwise accept a word outside $L_{p,q}$. The zero residual must nevertheless be represented by a final state. Hence one additional state is necessary, giving $p + q + 1$.

The same argument applies to the sign-separated model: theorem 8.8 forces at least $\max(p, q)$ states that occur on nonzero tails, none of which may be final, so a separate zero/final state is necessary. This gives $\max(p, q) + 1$.

For tightness, start from the corresponding right-endmarker construction and split the zero configuration into a fresh state Z carrying the same bottom marker Z_0 . Redirect every binary transition whose semantic target is residual zero to (Z, Z_0) , and define the two binary moves from (Z, Z_0) to be the existing encodings of residuals $-p$ and q . Declare only Z final. All other transition identities are unchanged, no ε -move is introduced, and every binary edge remains a single raw transition. $\square$

Thus the right endmarker saves exactly one control state in both scoped raw-conjugacy models. This acceptance-convention effect is separate from the stack-alphabet tradeoff: it changes the zero representation only and leaves the nonzero residual geometry unchanged.

Corollary 8.14 (Three-symbol raw configuration rectangles). *For every residual r with $\mu(r) = (x_r, y_r)$, the map*

$$\Psi_r^{\text{raw}}(i, j) = \zeta(r - pi + qj)$$

is a label-preserving isomorphism from $[0, x_r] \times [0, y_r]$ onto an ordinary DPDA shortest-path interval from $\zeta(r)$ to $\zeta(0)$. Every elementary square is a raw commuting diamond:

$$\boxed{\delta^*(\Psi_r^{\text{raw}}(i, j), 01) = \Psi_r^{\text{raw}}(i + 1, j + 1) = \delta^*(\Psi_r^{\text{raw}}(i, j), 10).}$$

Proof. Combine theorems 7.3 and 8.7. □

Example: $p = 3, q = 2, r = 4$ with no normalization and only three stack symbols. Here $p \geq q$ and

$$\begin{aligned} \zeta(4) &= (R_1, 0Z_0), & \zeta(1) &= (R_1, Z_0), & \zeta(-2) &= (R_0, 1Z_0), \\ \zeta(6) &= (R_0, 00Z_0), & \zeta(3) &= (R_0, 0Z_0), & \zeta(0) &= (R_0, Z_0). \end{aligned}$$

The boundary edge $1 \xrightarrow{0} -2$ is now the single rule

$$\delta(R_1, 0, Z_0) = (R_0, 1Z_0),$$

and the edge $-2 \xrightarrow{1} 0$ is the ordinary pop

$$\delta(R_0, 1, 1) = (R_0, \varepsilon).$$

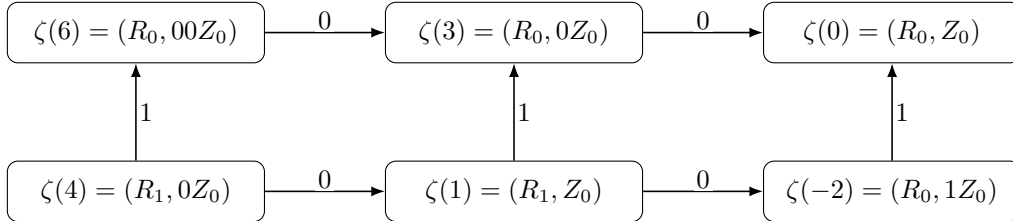


Figure 5: Raw configuration rectangle for $p = 3, q = 2, r = 4$ in the three-symbol boundary-compatible encoding. Every arrow is exactly one binary-input transition. The same rectangle required normalized macro-steps in the companion four-symbol coordinates of figure 4; changing the residual bases removes the boundary defect and also removes the extra sentinel X .

Corollary 8.15 (Coordinate-shift improvement). *The $p + q - 1$ raw defects of theorem 8.1 are properties of the fixed four-symbol homogeneous-tail coordinates, not a lower bound on stack-alphabet size. The boundary-compatible encoding achieves global raw residual-configuration conjugacy with one fewer stack symbol and the same control-state count:*

$$\boxed{(\max(p, q), 4, \text{normalized macro-steps}) \longrightarrow (\max(p, q), 3, \text{raw binary steps}).}$$

This correction sharpens the interpretation of the defect theorem. The missing information in the old coordinates was not inherently “one extra endpoint bit.” Rather, the chosen finite bases were misaligned with the sign-changing unary pops. Moving the finite bases to the adjacent residual classes lets the same state serve as the post-pop target, so the ordinary bottom marker Z_0 is sufficient. The construction is an explicit real-time-style one-counter realization. Corollary 8.6 gives an exact alphabet lower bound at the $\max(p, q)$ state budget inside the permanent-bottom raw-conjugacy model; no unrestricted recognition-level state or alphabet minimality is claimed.

9 Phase-local representation resources

The preceding results separate a finite local role from an unbounded magnitude. Before introducing coordinates, it is useful to isolate the elementary information principle behind the bound.

Lemma 9.1 (Local-role factorization). *Let τ be any finite semantic role on a class of configurations, and let*

$$\text{Obs}_\lambda(q, S) = (q, \text{pref}_\lambda^\perp(S)).$$

The role is exactly decodable from state plus the top λ stack symbols if and only if τ is constant on every fiber of Obs_λ .

Proof. A decoder implies constancy on observation fibers. Conversely, if the role is constant on each fiber, assign that common value to the corresponding observation; this defines a well-posed finite decoder on all reachable observations. $\square$

For the motivating DPDA, theorem 6.1 instantiates this principle with $\tau = \mathcal{G}$ and exact depth 2. The general resource model uses a richer bounded role: the signed canonical phase left after a coarse residual decomposition. The purpose of the following theorem is to count how much local observation capacity is required to keep that bounded phase visible while the remaining magnitude is unbounded. The results remain explicitly model-scoped.

Definition 9.2 (B -coordinate phase-local representation). Fix any integer $B \geq 1$ and write each residual as $r = s + Bz$, $0 \leq s < B$. We consider exact normalized representations satisfying:

- (i) the represented residual is recoverable without loss;
- (ii) each remainder s is assigned a specified set Q_s of control variants, with the Q_s disjoint for the global bound;
- (iii) the signed canonical phase is recoverable from the control state and the top λ work-stack symbols over an alphabet of size g .

For $r > 0$, define $\theta_B^+(r) = (+, y_r)$; for $r < 0$, define $\theta_B^-(r) = (-, x_r)$. When the remainder class contains $r = 0$, the zero residual is not counted in $H(B)$ below; including it would only strengthen the capacity lower bound.

Lemma 9.3 (Exact phase periods). *For fixed remainder s , the positive and negative phase periods are*

$$h_+(B) = \frac{p}{\gcd(B, p)}, \quad h_-(B) = \frac{q}{\gcd(B, q)}.$$

Hence the number of signed phases per remainder is

$$H(B) = \frac{p}{\gcd(B, p)} + \frac{q}{\gcd(B, q)}.$$

Each phase occurs at arbitrarily large stack depth in any exact finite-control, finite-stack-alphabet realization of theorem 9.2.

Proof. For $r = s + Bz > 0$, $qy_r \equiv -r \pmod{p}$; incrementing z changes y_r by the fixed step $-q^{-1}B$ modulo p . Since multiplication by the unit q^{-1} does not change the gcd with p , the additive order of this step is exactly $p/\gcd(B, p)$. The negative case uses $px_r \equiv r \pmod{q}$. Each phase recurs on an unbounded arithmetic progression of residuals. Exactness with finite control and finite alphabet prevents infinitely many distinct residuals from being represented at bounded stack depth. $\square$

Theorem 9.4 (Locality–control–alphabet capacity). *In the model of theorem 9.2, if $c_s = |Q_s|$, then for every remainder s ,*

$$\boxed{c_s g^\lambda \geq H(B)}.$$

If the remainder control sets are disjoint, then

$$\boxed{|Q| g^\lambda \geq BH(B)}.$$

Moreover

$$\boxed{BH(B)^2 \geq 4pq},$$

so

$$\boxed{|Q| g^\lambda \geq 2\sqrt{pqB}}.$$

Proof. By theorem 9.3, all $H(B)$ signed phases occur at depth at least λ . A local observation has at most $c_s g^\lambda$ values, so exact phase decoding requires $c_s g^\lambda \geq H(B)$. Summing $c_s g^\lambda \geq H(B)$ over the B remainder classes gives $(\sum_s c_s) g^\lambda \geq BH(B)$; disjointness implies $|Q| \geq \sum_s c_s$, yielding the global bound. Let $a = \gcd(B, p)$ and $b = \gcd(B, q)$. Since $\gcd(p, q) = 1$, a and b are coprime divisors of B , hence $ab \leq B$. Therefore

$$h_+(B)h_-(B) = \frac{pq}{ab} \geq \frac{pq}{B}.$$

AM–GM yields $H(B)^2 \geq 4pq/B$, which gives the remaining inequality. If a chosen coordinate architecture additionally satisfies $B \geq M := \max(p, q)$, then $|Q| g^\lambda \geq 2\sqrt{pqM}$ follows immediately. $\square$

In particular, in the common regime $B \geq M := \max(p, q)$,

$$\boxed{|Q| g^\lambda \geq 2\sqrt{pqM}}.$$

The phase-capacity theorem itself, however, does not require $B \geq M$.

Corollary 9.5 (An infinite family). *Let*

$$p = k^2, \quad q = k^2 - 1, \quad B = k(k + 1), \quad k \geq 2.$$

Then

$$H(B) = 2k - 1$$

and

$$\boxed{|Q| g^\lambda \geq k(k + 1)(2k - 1) = 2k^3 + k^2 - k}.$$

With $M = k^2$,

$$\boxed{|Q| g^\lambda \geq 2M^{3/2} + M - \sqrt{M} = \Omega(M^{3/2})}.$$

If $|Q| \leq CB$, then $g^\lambda \geq (2k - 1)/C$; in particular fixed $\lambda = 2$ forces $g = \Omega(M^{1/4})$, while fixed $g > 1$ forces $\lambda = \Omega(\log M)$.

Proof. Here $\gcd(B, p) = k$ and $\gcd(B, q) = k + 1$, so $h_+ = k$ and $h_- = k - 1$. Substitute into theorem 9.4 and simplify. $\square$

The family asymptotically approaches the arithmetic conservation envelope: the ratio

$$\frac{BH(B)}{2\sqrt{pqB}} = \frac{2k - 1}{2\sqrt{k(k - 1)}} = \sqrt{1 + \frac{1}{4k(k - 1)}} \rightarrow 1.$$

This is saturation of the arithmetic lower-bound mechanism only; it is not a matching DPDA upper bound or a complete Pareto frontier.

10 Discussion, limitations, and open problems

The original DPDA is useful precisely because it exposes structure that recognition alone hides. Completion, cancellation, and bounded visibility are three views of one minimum demand. Completion materializes the demand in a canonical ordering; cancellation consumes part of it online; the first two stack symbols reveal the qualitative shape of the remaining shortest-path geometry without revealing its unbounded scale.

The general rectangle separates a rigid Parikh demand from flexible geodesic ordering. The demand (x_r, y_r) is unique, whereas its shortest orderings form $\binom{x_r + y_r}{x_r}$ monotone paths. Across all residuals, the global geodesic-word count has its first obstruction exactly at the fundamental cycle length $p + q$: every shorter binary word is geodesic, while the later count is a pair of fixed binomial tails and hence has exact polynomial degree $\max(p, q) - 1$ for every nontrivial coprime pair. Canonical completion is one extremal boundary; the opposite block order is the other; and $\Xi(r) = x_r y_r$ counts both commuting squares and maximal adjacent normalization distance. The four-symbol residual-configuration conjugacy shows that this rectangle is not merely arithmetic: its vertices are actual normalized stable configurations and its edges are deterministic one-symbol macro-steps. The raw-defect theorem identifies exactly where those particular coordinates need normalization. The three-symbol construction then shows that the defect is coordinate-relative: aligning the finite boundary bases with the sign-changing pop makes the entire residual graph—hence every rectangle—literal under ordinary one-input transitions. Global Parikh synchronization therefore holds both modulo normalization in the companion four-symbol coordinates and literally in the three-symbol boundary-compatible machine. The one-work-symbol theorem and the homogeneous-tail frontier make the accompanying descriptive tradeoff exact throughout that scoped class: one work symbol requires $p + q$ states, while two work symbols already permit the exact minimum $\max(p, q)$ and additional homogeneous-tail symbols give no further state reduction. At the ratios $(2, 1)$ and $(3, 1)$, the respective $\max(p, q)$ lower bounds survive even when the homogeneous-tail hypothesis is removed entirely. A computer-assisted finite certificate establishes the same exact three-state bound for the first genuinely two-sided ratio $(3, 2)$ (and symmetrically $(2, 3)$).

What is not claimed. The resource inequalities in section 9 apply only to the explicitly defined exact, remainder-separated, phase-local B -coordinate architecture. They do not imply an unrestricted state lower bound for all DPDAs recognizing $L_{p,q}$ or a complete state-alphabet-visibility Pareto frontier. Separately, theorems 8.5, 8.6 and 8.9 give exact lower bounds only for explicitly stated permanent-bottom raw-conjugacy classes, not for arbitrary recognizers. A matching upper bound would additionally require a standard-DPDA implementation theorem showing that a compressed local window can be maintained using legal top-stack transitions; no such general compiler is proved here.

External novelty boundary. The mathematical statements above are proved directly, but several ingredients have close prior art. Chistikov et al. study tight shortest-path bounds in one-counter systems [4]. More directly for pushdown graphs, Carayol and Serre mark exactly those outgoing edges that belong to some shortest path to a final vertex [8]; our geodesic first-action set is therefore a specialized shortest-edge marking, not a new general notion. Repeated commutation of independent adjacent letters and normal-form selection are standard in trace theory [9], and geodesic normal forms occur in other algebraic/formal-language settings [6]. Store-language analysis is classical [5]; stack-alphabet hierarchies and exact state/stack-symbol descriptive bounds also occur in other pushdown models [7, 10].

Accordingly, the submission claim is the exact family-specific synthesis: the concrete configuration ladder and its commuting diamonds in the motivating DPDA, the completion–cancellation

boundary interpretation, the tight depth-two decoder for its shortest-edge role, the residual rectangle package for $L_{p,q}$, the exact one-work-symbol raw-conjugacy state bound and three-symbol fixed-budget realization, and the scoped phase-local capacity inequality. We therefore state the contribution at the level of these exact proved results and avoid an absolute priority claim for the combined theorem chain.

Open problems. Three directions appear particularly natural. First, determine the unrestricted permanent-bottom raw-conjugacy state–alphabet frontier outside the complete homogeneous-tail subclass. The present paper proves the full homogeneous-tail frontier and removes that structural restriction for $(2, 1)$, $(3, 1)$, and $(3, 2)$ (and their symmetric partners), but the arbitrary three-symbol state minimum for general coprime (p, q) remains open. Second, characterize when a standard DPDA can maintain an abstract top- λ semantic window with bounded overhead and seek matching upper bounds for the phase-local resource family. Third, extend the completion/locality viewpoint to higher-dimensional counting constraints, where rectangles may be replaced by higher-dimensional lattice intervals.

11 Conclusion

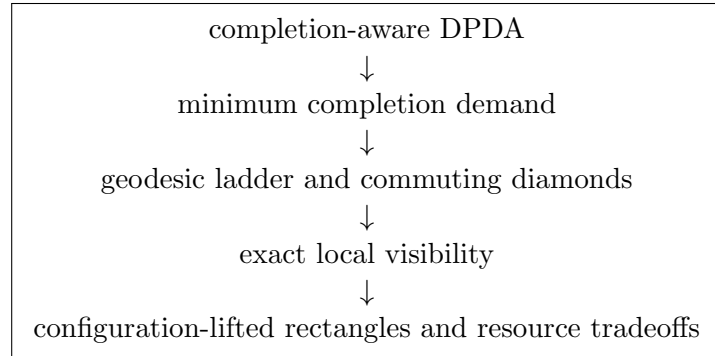
A concrete DPDA for $L_{2,1}$ contains more structure than is needed for recognition. Its left-state cancellation loop consumes one unit of a unique minimum completion demand. Its reachable left branch realizes every shortest ordering of that demand as an actual computation, producing an exact ladder of commuting diamonds. End-of-input completion and maximal cancellation are the two boundary paths of this ladder. Thus completion and cancellation are not independent mechanisms: they are different schedules for consuming one minimum demand.

The same geometry explains the original two-symbol stack observation. State plus two stack symbols recover the complete local geodesic-action set, while no fixed depth recovers the unbounded numerical demand. The machine therefore exhibits a clean separation between bounded qualitative geometry and unbounded quantitative magnitude.

For general $L_{p,q}$, the ladder becomes the rectangle $[0, x_r] \times [0, y_r]$. Every monotone path is a shortest completion, the number of paths is $\binom{x_r + y_r}{x_r}$, and the number of commuting squares is $x_r y_r$. Every binary word shorter than $p + q$ is geodesic; the first failures are the fundamental balanced-cycle words at length $p + q$, after which the exact geodesic-word count has polynomial growth of degree $\max(p, q) - 1$ for every nontrivial coprime pair. Geodesic forks are classified by divisibility of the residual, and the canonical completion exposes exactly the locally optimal symbol support with exact visibility depth $\max(p, q)$. In the compact four-symbol paired-sign DPDA, the residual graph is label-isomorphic to the normalized stable configuration graph, with exactly $p + q - 1$ raw boundary defects for the fixed homogeneous-tail encoding. A boundary-compatible change of residual coordinates removes those defects altogether and yields a three-symbol, ε -free right-endmarker DPDA with the same $\max(p, q)$ control states whose ordinary binary transition graph is itself label-isomorphic to the residual graph. Hence every residual rectangle lifts not only to normalized configurations but also to a raw three-symbol DPDA configuration rectangle. In the asymptotically homogeneous-tail permanent-bottom raw-conjugacy model, the state–alphabet frontier is exact: one work symbol forces $p + q$ states, while any work alphabet with at least two symbols has exact minimum $\max(p, q)$. The three-symbol total alphabet (two work symbols plus the bottom marker) is therefore minimal at the $\max(p, q)$ state budget; for $(2, 1)$, $(3, 1)$, and $(3, 2)$ the respective $\max(p, q)$ lower bounds are exact even for arbitrary permanent-bottom three-symbol encodings.

Finally, the phase-local model yields a scoped three-resource capacity theorem. The structural

chain is



The central lesson is that the original stack manipulations are not merely implementation details. They expose a geometry of shortest completion that survives beyond the original 2:1 automaton.

A Result status and scope

Result	Status	Scope
Reachable-store normal form	proved	motivating DPDA
State-potential invariant	proved	motivating DPDA
Minimum-demand geodesic descent	proved	completion residuals
Exact geodesic ladder and diamonds	proved	reachable left branch
Completion-cancellation duality	proved	reachable left branch
Exact depth-two geodesic locality	proved, tight	motivating DPDA
Full demand not fixed-depth local	proved	motivating representation
General shortest-completion rectangle	proved	all coprime p, q
Exact geodesic-word growth	proved	all coprime p, q / first obstruction at $p + q$
Normalized residual-configuration conjugacy	proved	four-symbol paired-sign DPDA / macro-steps
Configuration-level rectangle lift	proved	normalized stable configuration graph
Exact four-symbol raw-defect set	proved, tight for fixed encoding	$p + q - 1$ boundary edges
One-work-symbol raw-conjugacy state complexity	proved, tight	permanent-bottom model / $ Q = p + q, \Gamma = 2$
Alphabet minimality at $ Q = \max(p, q)$	proved, tight	permanent-bottom raw-conjugacy model / $ \Gamma = 3$
Three-symbol raw residual-configuration conjugacy	proved	ε -free right-endmarker DPDA / $ Q = \max(p, q)$
Sign-separated homogeneous-tail state complexity	proved, tight	permanent-bottom raw-conjugacy subclass / $ Q = \max(p, q)$
Homogeneous-tail alphabet frontier	proved, tight	$k = 1$: $p + q$ states; $k \geq 2$: $\max(p, q)$ states
Three-symbol state complexity for $(2, 1)$	proved, tight	arbitrary permanent-bottom raw encoding / $ Q = 2$
Three-symbol state complexity for $(3, 1)$	proved, tight	arbitrary permanent-bottom raw encoding / $ Q = 3$
Three-symbol state complexity for $(3, 2)$	computer-assisted, tight	arbitrary permanent-bottom raw encoding / $ Q = 3$
No-endmarker acceptance cost	proved, tight	scoped raw-conjugacy models / exactly one extra state
Raw configuration-level rectangle lift	proved	three-symbol ordinary transition graph
Exact fork classification	proved	all coprime p, q
Canonical action visibility = $\max(p, q)$	proved, tight	canonical representation
B -phase periods and capacity	proved	phase-local B -coordinate model
Three-resource lower bound	proved	same model
Executable consistency audit	passed	non-proof evidence
Matching general upper bound	open	—
Unrestricted DPDA Pareto frontier	open	—
Unrestricted state optimality	open	—

B Executable consistency checks

All results above except the explicitly labeled computer-assisted $(3, 2)$ state lower bound are proved symbolically. A separate Python script serves as an adversarial consistency check for those symbolic results and, for theorem 8.12, as the exhaustive finite certificate after the proof has reduced the problem to a bounded search space. The audit verifies all solid core-transition potential changes, completion materialization on deep reachable families, and the exact motivating ladder for $0 \leq k \leq 150$. It independently brute-forces the minimum-demand problem on 11,011 small coprime/residual cases and then performs broad closed-form arithmetic checks on 1,548,547 additional residual cases. For the configuration-lift theorem it checks 667,110 compact normalized input macro-steps and explicitly enumerates 1,859,269 normalized rectangle vertices containing 1,286,746 commuting diamonds. For the general geodesic count it independently checks 67,155 coprime-parameter/length cases against a direct Parikh-pair enumeration, including the first balanced-cycle obstruction at length $p + q$. For the raw-conjugacy extension it checks 1,111,110 immediate four-symbol edges against the exact defect classification, 1,111,110 one-work-symbol raw updates, 1,111,110 three-symbol one-step residual updates, and an additional 1,111,110 split-zero no-endmarker update checks for each of the two- and three-symbol constructions. For the unrestricted one-state three-symbol obstruction at $(p, q) = (2, 1)$, an independent finite audit exhausts all 50,625 bounded rule systems in the positive-minimum reduction and all 2,126,250 bounded rule systems in the zero-minimum reduction; no candidate survives the required inverse identities. For the arbitrary two-state obstruction at $(p, q) = (3, 1)$, a separate prefix-synthesis audit explores 98,610 positive-minimum branch attempts and 258,777 zero-minimum branch attempts, reproduces exactly the four and two symmetry-related terminal prefixes displayed by the finite reduction, and confirms that none extends one further step. For the computer-assisted $(3, 2)$ lower bound, the finite certificate explores 1,130,430 positive-minimum and 1,696,120 zero-minimum branch attempts, asserts the exact survivor vectors stated in theorem 8.12, and terminates with zero candidates in both cases. It also checks 61,455 four-symbol raw-rectangle classifications and 731,091 three-symbol raw rectangle vertices containing 474,684 raw commuting diamonds, together with exact visibility witnesses, phase periods for values of B both below and above $\max(p, q)$, coordinate conservation, and the infinite-family formulas. Across the full run, the script performs 90,764,832 assertion-level checks; all pass.

Validation file: `second_paper_validation_suite.py`. It is distributed separately with the submission.

Project repository: <https://github.com/alperenbutun/dpda-ones-double-zeros>.

Outside theorem 8.12, its role is to detect transcription and boundary-case errors rather than to replace a theorem proof. For that one computer-assisted theorem, the symbolic argument establishes finiteness and the legal pruning bounds, while the distributed script exhausts the remaining bounded cases.

Statements and Declarations

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